Leptonic Compactness and the Generalized n-Dimensional Velocity Law

A formal report on the `TheoreticalFramework` development

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Status: **all checks passed**.
Lean: `4.33.1`; **Mathlib:** prebuilt at `/opt/mathlib`.
Sources: 5 modules, 97 theorems, 42 definitions, 3 structures.
Axioms used across the development: `Classical.choice`, `Quot.sound`, `propext`.

Summary

Part of the muon's anomalous magnetic moment is attributed, in the framework under study, to an internal geometric compactness of massive leptons with effective radius $r_\ell = \kappa/m_\ell$. Under the framework's plate ansatz (S proportional to m^2, separation $d = r_\ell$) the Casimir energy of the internal geometry is quintic in the mass, so the amplification between two leptons is the fifth power of their mass ratio: $E_{\mathrm{C},\mu}/E_{\mathrm{C},e} \approx (207)^5 \approx 4 \times 10^{11}$ and $E_{\mathrm{C},\tau}/E_{\mathrm{C},e} \approx (3477)^5 \approx 5.1 \times 10^{17}$. Within the generalized n-dimensional kinematics with $n = 5$, the same quintic dependence fixes the ejection-velocity law $v_f^2 = v_2^2\,(m_\ell/m_0)^{n-2}$, hence amplification factors $(207)^3 \approx 8.9 \times 10^6$ and $(3477)^3 \approx 4.2 \times 10^{10}$. This report does three things. First, it records the derivation of every scaling law as a Lean theorem, so that the fifth-power and cube appear as consequences rather than assertions, and so that the parameter-freedom of the ratios is visible. Second, it recomputes every physical constant independently in Python and compares it with the interval Lean states — including the three places where the source text's arithmetic does not survive that comparison. Third, it separates what the development proves from what the framework assumes and from what would need experiment.

1 Scope, and what is and is not proved

The development is deliberately layered, and the layers should not be confused.

(A) Proved mathematics. The scaling laws, the amplification ratios, the exact arithmetic of the ejecta, the algebra of the regime structure, the continuity of the velocity map, and the thrust–power identity. These are Lean theorems with no hypotheses beyond positivity of the masses and ratios, and they hold whatever one thinks of the physics.

(B) Framework postulates. Two assumptions are *recorded* rather than derived, and the formalization makes that visible instead of hiding it. The geometric ansatz (S proportional to m^2 with $d = r_\ell$, and $r_\ell = \kappa/m_\ell$) is a structure (`CasimirAnsatz`) carrying a free constant κ and a free area coefficient; the theorem that matters is that the *ratio* does not depend on either of them (`CasimirAnsatz.energy_ratio_parameter_free`). The velocity law $v_f^2 = v_2^2\,(m_\ell/m_0)^{n-2}$ is likewise a postulate of the framework's generalized kinematics, and the development proves its consequences (`velocity_law`), not its truth.

(C) Empirical claims. That any of this describes the muon's actual internal structure, that muon–muon collisions eject particles at these speeds, and that a thruster can be built, are questions for experiment. The development is explicit about the most important of these: §7 proves that the geometry fixes the exhaust velocity and the thrust-to-power ratio but says nothing about absolute thrust, and the claim that is left open is recorded as a proposition (`ThrustOptimizationConjecture`) that is never asserted.

A tension internal to the source. The framework attributes part of a_μ to a term scaling as m^5, whereas the measured generation-dependence of the anomalous moment goes as roughly $(m_\mu/m_e)^2 \approx 4.3 \times 10^4$, a factor $\approx 10^7$ smaller than $(207)^5$. The attached notes say so themselves. Nothing in this report resolves that: the formalization establishes that the framework's own arithmetic is correct, which is a different statement from the framework being right. A Lean proof of a scaling law is not evidence for the postulate that produces it.

2 Core derivation: compactness and the fifth power

Write the effective radius as $r_\ell = \kappa/m_\ell$ and the Casimir energy of the internal cavity as $E_{\mathsf{C}} \propto -S/d^3$ with $d = r_\ell$. Three theorems carry the derivation.

1. `casimirEnergy_eq_quintic`: for $\kappa \neq 0$ and $m \neq 0$,
$$E_{\mathsf{C}}(\kappa, m) \;=\; \frac{m^2}{(\kappa/m)^3} \;=\; \frac{m^5}{\kappa^3}.$$
2. `casimirEnergy_ratio`: for non-zero masses,
$$\frac{E_{\mathsf{C}}(\kappa, m_1)}{E_{\mathsf{C}}(\kappa, m_2)} \;=\; \left(\frac{m_1}{m_2}\right)^5.$$
3. `casimirEnergy_ratio_radii`: the same number read geometrically,
$$\frac{E_{\mathsf{C}}(\kappa, m_1)}{E_{\mathsf{C}}(\kappa, m_2)} \;=\; \left(\frac{r_2}{r_1}\right)^5,$$
so "compactness" and "vacuum coupling" are one quantity, not two.

The constant κ cancels in the ratio. That is the whole reason the framework can issue a number at all, and the formalization isolates it: in `CasimirAnsatz` the ratio is proved independent of κ, of the area coefficient, and of the overall Casimir prefactor, and the sign convention (the physical energy is negative, i.e. attractive) is shown not to affect it (`signedEnergy_ratio`).

The effective radius is strictly decreasing in the mass (`effRadius_strictAnti`), which is the compactness statement in the form the rest of the development uses.

3 Casimir amplification: the numbers

With $m_\mu/m_e = 207$ and $m_\tau/m_e = 3477$ — both checked against the tabulated masses to better than a quarter and a half unit respectively (`massRatioMuon_consistent`, `massRatioTauon_consistent`) — the fifth-power law gives the following. "Lean" is the interval the theorem …`_order` states; "recomputed" is this script's independent evaluation; "published" is the source text.

Quantity	Lean interval	Recomputed	Published
$E_{\mathsf{C},\mu}/E_{\mathsf{C},e}$	$[3.8\times10^{11},\ 4\times10^{11}]$	3.800596×10^{11}	4×10^{11}
$E_{\mathsf{C},\tau}/E_{\mathsf{C},e}$	$[5.08\times10^{17},\ 5.1\times10^{17}]$	5.081869×10^{17}	5.1×10^{17}
$E_{\mathsf{C},\tau}/E_{\mathsf{C},\mu}$	$[1.3\times10^{6},\ 1.34\times10^{6}]$	1.337124×10^{6}	1.3×10^{6}

Two further structural facts are proved rather than computed. The amplification is monotone in the mass ratio (`casimir_amplification_hierarchy`: $1 < (207)^5 < (3477)^5$), so the lepton generation hierarchy is a strict hierarchy of vacuum couplings. And the equivalent magnetic field B, defined by $B^2 \propto E_{\mathsf{C}}/V$, carries exponent 3 under the source's volume ansatz V proportional to Sd^3, and exponent 2 under the literal plate volume $V = Sd$. Both are proved (`field_amplification_source`, `field_amplification_plate`). The exponent 3 is the one the framework uses, and it is the same $3 = n - 2$ that appears in the velocity law — which is worth stating precisely, because the coincidence is the framework's only argument for $n = 5$ beyond the quintic mass dependence itself.

4 The generalized n-dimensional framework

The generalized dimension is not tuned: it is read off the quintic mass dependence of the anomalous-moment term. In the development,

```
casimirMassExponent   = 5
generalizedDimension  = casimirMassExponent
velocityExponent      = generalizedDimension - 2  = 3
```

so that `generalizedDimension_eq_five` and `velocityExponent_eq_three` are definitional, and the exponent of the velocity law is a consequence of the Casimir exponent rather than a choice made independently of it.

The link to $g-2$ is made explicit rather than left in prose. The framework's decomposition $a_\ell = a_{\mathsf{QED}} + a_{\mathsf{EW+had}} + a_{\mathsf{str}}$ is the definition `anomalousMoment`, its structural part is `structuralTerm` $= C\,m^n$, and `structuralTerm_quintic` records that this is $C\,m^5$ while `structuralTerm_ratio` records that the structural term amplifies by $(m_1/m_2)^5$ — the same number as the Casimir amplification. That identity of exponents is the whole basis for reading $n = 5$ off the anomalous moment, so it is worth having as a theorem rather than a remark.

The law is stated for arbitrary n and then specialized. With $r = m_\ell/m_0$,

$$v_f \;=\; \sqrt{v_2^2\, r^{\,n-2}}, \qquad \text{so} \qquad v_f^2 \;=\; v_2^2\, r^{\,n-2}$$

(`velocity_law`), and for $v_2 \geq 0$ this is the paper's multiplicative form $v_f = v_2\sqrt{r^{\,n-2}}$ (`generalisedVelocity_eq_mul`). At $n = 5$ both become the cube (`velocity_law_five`). The map is monotone in r and in v_2, vanishes at $v_2 = 0$, and is continuous in each variable.

The framework's generalized units enter only as exponent bookkeeping, and that bookkeeping is proved exactly: with $[v_n] = \mathsf{m}^{2/n}/\mathsf{s}$ the mass exponent balances as $n \cdot (2/n) = 2$ (`generalised_unit_exponents`) and the time exponent as $(n-2) + 2 = n$ (`time_exponent_balance`). The substitution rule $v_n(q\,dt_0)^{n-2} = v_2^2 q^{n-2}$ that links the two regimes is `scale_substitution`.

5 Velocity scaling and amplification

The amplification factors are exact integers: $207^3 = 8\,869\,743$ and $3477^3 = 42\,035\,292\,333$ (`muonAmplification_eq`, `tauonAmplification_eq`), i.e. $\approx 8.9 \times 10^6$ and $\approx 4.2 \times 10^{10}$. Their ratio is $(3477/207)^3 \approx 4.74 \times 10^3$.

Because v_f^2 is an exact rational, every velocity is bounded without any rounding step: the development computes v_f^2 exactly and closes the interval with `generalisedVelocity_bounds`.

Quantity	Lean interval (m/s unless stated)	Recomputed	Published
muonic ceiling v_f	$[2.0682 \times 10^7,\ 2.06821 \times 10^7]$	2.068202×10^7	$\approx 20\,700$ km/s
muonic ceiling (km/s)	[20,680, 20,700]	20,682	—
muonic I_{sp} (s)	$[2.108 \times 10^6,\ 2.109 \times 10^6]$	2.108979×10^6	$\approx 2 \times 10^6$
$I_{sp}/I_{sp,Raptor}$	[5,800, 5,820]	5,810	≈ 6000
tauonic ceiling v_f	$[4.2403 \times 10^{12},\ 4.2404 \times 10^{12}]$	4.240334×10^{12}	1.34×10^{12}
tauonic at entry	$[1.4237 \times 10^9,\ 1.4239 \times 10^9]$	1.423785×10^9	1.42×10^9
tauonic at median	$[2.8475 \times 10^{10},\ 2.8476 \times 10^{10}]$	2.847571×10^{10}	9.0×10^{10}

The ceiling is at the muonic collision maximum $v_2 = 25\,000$ km/h $= 62500/9 \approx 6\,944.4$ m/s (`collisionVelocityMuonMax_eq`).

The law over the whole input range

Values computed from the verified law $v_f^2 = v_2^2\, r^3$; only the endpoints are bounded inside Lean, the rest are consequences.

v_2 (km/h)	v_2 (m/s)	v_f muonic (m/s)	v_f tauonic (m/s)
0	0	0	0
2,500	694	2.068202×10^6	1.423785×10^8
5,000	1,389	4.136405×10^6	2.847571×10^8
10,000	2,778	8.272809×10^6	5.695142×10^8
25,000	6,944	2.068202×10^7	1.423785×10^9
50,000	13,889	4.136405×10^7	2.847571×10^9
250,000	69,444	2.068202×10^8	1.423785×10^{10}
1,000,000	2.777778×10^5	8.272809×10^8	5.695142×10^{10}

Rows above $25\,000$ km/h lie outside the muonic input window and are tabulated only to show the law's behaviour over three decades; they are not outputs of a regime. The muonic column above $\approx 10^5$ km/h is already superluminal, which is one reason the muonic window stops where it does.

Where the source text diverges

Three published figures were checked against the exact law. Two do not survive, and the pattern is the same in both: the source computes v_f^2 correctly and then takes its square root with the decimal exponent misplaced, which produces a factor $\sqrt{10} \approx 3.16$ on v_f whenever the exponent of v_f^2 is odd.

Quantity	Published	Verified	Ratio	Diagnosis
tauonic ceiling	1.34×10^{12}	4.2403×10^{12}	3.1644	$\sqrt{10}$ slip
tauonic at median	9.0×10^{10}	2.8476×10^{10}	0.3164	same slip, other direction
tauonic at entry	1.42×10^9	1.4238×10^9	1.003	correct (even exponent)

The slip is not asserted but localised: `source_tauon_max_velocity_underestimates` proves that the published figure squared is a factor in $[10.01, 10.02]$ below the verified one, and `source_tauon_max_velocity_factor` that the figure itself is low by a factor in $[3.164, 3.165]$. The source's own intermediate, $v_f^2 \approx 1.80 \times 10^{25}$ m²/s², is right; only its root is wrong. Both values are kept in the development (`sourceTauonMaxVelocity`) so that nothing is silently corrected.

The speed limit

The muonic ceiling is subluminal (`muonMaxEjection_subluminal`: $2.07 \times 10^7 < c$), and so is the muonic collision window. The tauonic ceiling is not (`tauonMaxEjection_superluminal`: $c < 4.24 \times 10^{12}$ m/s). The tauonic *input* window is still subluminal (`tauonicInputMax_subluminal`), so the framework's rejection of the Lorentz factor is required only to interpret the output. This is the sharpest formal statement of the framework's distance from standard kinematics, and it is worth being explicit: a superluminal exhaust is not a technical difficulty to be engineered around, it is a consequence of the law as written.

6 The two regimes, and what continuity actually says

Both regimes are instances of one structure, `PropulsiveRegime`, carrying a mass ratio, an input window and the law-induced output. Everything proved of the structure therefore applies to both.

	muonic	tauonic
Colliding leptons	μ^-/μ^+	τ^-/τ^+
Mass ratio m/m_e	207	3477
Amplification $(m/m_e)^3$	8.87×10^6	4.20×10^{10}
Casimir amplification	3.80×10^{11}	5.08×10^{17}
Input window v_2	$]0,\ 25\,000$ km/h$]$	$]25\,000$ km/h$,\ v_f^{\max,\mu}]$
Output ceiling	$20\,682$ km/s	4.2403×10^{12} m/s

Continuity, stated correctly. The source text says that "the entry velocity of the tauonic domain coincides exactly with the maximum output velocity of the muonic domain". Those are two different coincidences, and the formalization separates them, because conflating them is how a boundary argument goes wrong:

1. **Adjacency** (`windows_adjacent`): the tauonic window begins at $25\,000$ km/h, which is the muonic *input* ceiling.
2. **The chain closes** (`chain_closes`): the tauonic window's *upper* end is the muonic regime's maximum *output*, $20\,682$ km/s. So the second regime's input ceiling is the first regime's output ceiling — true of the upper end, not the entry.

Together with `regimes_tile` (the two windows cover one interval, with no gap and no overlap) and `tauonic_window_nested_in_output` (the tauonic window sits inside the muonic output range), this is the exact content of the nested-regime claim. `regime_continuity` collects the four facts and the continuity of the velocity map in v_2 in one statement.

The boundary itself is a named object, `domainBoundary`, with `domainBoundary_eq_muonic_output` (it *is* the muonic output ceiling) and

domainBoundary_km_per_s (it lies in $[20\,680, 20\,700]$ km/s, i.e. the source's $20\,700$ to the precision of the print). It is subluminal (domainBoundary_subluminal).

Taking the unrounded value $v_f^{\mathrm{max},\mu}$ as the tauonic input ceiling rather than the printed $20\,700$ km/s is what makes the match exact rather than approximate: the theorem chain_closes is then true by construction, whereas with the rounded figure it would hold only to three significant digits. That choice is recorded in the definition of tauonicInputMax and justified in its docstring.

Structural algebra. The amplification $r \mapsto r^{n-2}$ is a multiplicative character (amplification_mul), so the tauonic factor factors through the muonic one (tauon_amplification_factors): $3477^3 = 207^3 \cdot (3477/207)^3$. This is why the regimes compose: a two-stage cascade multiplies, and the ordering of the regimes is forced by the lepton mass hierarchy rather than chosen (regimes_strictly_ordered).

7 Thrust: what the framework fixes, and the question it leaves open

The kinematics fix the exhaust velocity. Momentum and energy balance then fix the thrust-to-power ratio exactly, and nothing else:

$$T = \dot{m}\, v_e, \qquad P = \tfrac{1}{2}\dot{m}\, v_e^2 \quad \Longrightarrow \quad \frac{T}{P} = \frac{2}{v_e}$$

proved as thrust_to_power for an arbitrary ThrusterDesign, and as thrust_eq_power: $T = 2P/v_e$.

This has a consequence the source text does not mention and which the development makes unavoidable. At the muonic exhaust velocity, $T/P \approx 9.67 \times 10^{-8}$ N/W (muonic_thrust_to_power_bounds), which is a factor $\approx 5.8 \times 10^3$ *worse* than a Raptor engine (muonic_thrust_to_power_shortfall). The specific-impulse gain and the thrust-per-watt loss are the same number, because both are v_e: raising I_{sp} from 363 s to 2.1×10^6 s raises it by 5.8×10^3 and lowers thrust per watt by 5.8×10^3. Any muonic thruster trades one for the other; there is no regime in which the law gives both.

What the framework does *not* fix is absolute thrust. This is proved, not asserted: thrust_not_determined shows that for any positive thrust T there is a design at the muonic exhaust velocity delivering exactly T — thrust is bought with mass flow and power. Consequently the engineering question is a question about power budgets, and is recorded as a proposition

```
ThrustOptimizationConjecture T P  :=
  exists a design with exhaust velocity the muonic ceiling,
  thrust T, and jet power at most P
```

which the development never asserts. It is shown to be non-vacuous (thrust_optimization_satisfiable) and to be settled by a power budget rather than by the geometry (thrust_optimization_is_empirical). That is the precise sense in which "empirical thrust optimisations remain unproven": the framework's contribution is the hyperbola $T = 2P/v_e$, and everything beyond it is engineering.

8 Verification

Overall: pass — sources elaborate, every constant agrees, no forbidden token, no non-standard axiom

1. Hygiene. The sources are scanned for `sorry`, `admit`, `axiom` and `native_decide` before anything is compiled; a hit aborts the build. Hits: **none**.

2. Elaboration. Each module is elaborated with `lean` in dependency order. Total elaboration time: 13.06 s.

Module	Status	Time	Result
`Basic`	elaborated	2.13 s	ok
`Leptons`	elaborated	2.12 s	ok
`Dimensions`	elaborated	2.08 s	ok
`Velocity`	elaborated	2.43 s	ok
`Regimes`	elaborated	2.42 s	ok
`TheoreticalFramework`	elaborated	1.88 s	ok

3. Axiom audit. `#print axioms` is run on every theorem and definition. Grouped by the set of axioms each declaration depends on:

Axioms	Declarations
none	6
`propext`, `Classical.choice`, `Quot.sound`	133

None of these is `sorryAx`, and no declaration carries a framework-specific axiom: the `propext` / `Classical.choice` / `Quot.sound` triple is Lean's standard classical base, used by the decision procedures, not by the mathematics of the framework.

4. Independent recomputation. Every numerical claim is recomputed in Python from its definition — exact integers and `Fraction` where the quantity is rational — and compared with the interval or numeral that Lean actually states.

Theorem	Quantity	Lean	Recomputed	
`collisionVelocityMuonMax_bounds`	muonic collision ceiling v_2, m/s	$[6{,}944,\ 6{,}945]$	$6{,}944$	ok
`casimir_amplification_muon_order`	casimir amplification $(m_\mu/m_e)^5$	$[3.8\times10^{11},\ 4\times10^{11}]$	3.800596×10^{11}	ok
`casimir_amplification_tauon_order`	casimir amplification $(m_\tau/m_e)^5$	$[5.08\times10^{17},\ 5.1\times10^{17}]$	5.081869×10^{17}	ok
`casimir_amplification_tauon_over_muon`	$(m_\tau/m_\mu)^5$	$[1.3\times10^{6},\ 1.34\times10^{6}]$	1.337124×10^{6}	ok
`muonAmplification_order`	velocity amplification 207^3	$[8.8\times10^{6},\ 8.9\times10^{6}]$	8.869743×10^{6}	ok
`tauonAmplification_order`	velocity amplification 3477^3	$[4.2\times10^{10},\ 4.21\times10^{10}]$	4.203529×10^{10}	ok
`tauon_over_muon_amplification`	$3477^3/207^3$	$[4{,}730,\ 4{,}740]$	$4{,}739$	ok
`muonMaxEjection_bounds`	muonic ejection ceiling, m/s	$[2.0682\times10^{7},\ 2.06821\times10^{7}]$	2.068202×10^{7}	ok
`muonMaxEjection_km_per_s`	muonic ejection ceiling, km/s	$[20{,}680,\ 20{,}700]$	$20{,}682$	ok

Theorem	Quantity	Lean	Recomputed	
muonSpecificImpulse_bounds	muonic specific impulse, s	$[2.108 \times 10^6,\ 2.109 \times 10^6]$	2.108979×10^6	ok
muonImpulse_over_raptor	Isp(muonic)/Isp(Raptor)	$[5{,}800,\ 5{,}820]$	$5{,}810$	ok
tauonMaxEjection_bounds	tauonic ejection ceiling, m/s	$[4.2403 \times 10^{12},\ 4.2404 \times 10^{12}]$	4.240334×10^{12}	ok
tauonEntryEjection_bounds	tauonic ejection at entry, m/s	$[1.4237 \times 10^9,\ 1.4239 \times 10^9]$	1.423785×10^9	ok
tauonMedianEjection_bounds	tauonic ejection at median input, m/s	$[2.8475 \times 10^{10},\ 2.8476 \times 10^{10}]$	2.847571×10^{10}	ok
source_tauon_max_velocity_underestimates	(verified/source)² at the tauonic ceiling	$[10.01,\ 10.02]$	10.0136	ok
source_tauon_max_velocity_factor	verified/source at the tauonic ceiling	$[3.164,\ 3.165]$	3.1644	ok
muonic_thrust_to_power_bounds	muonic thrust per watt, N/W	$[9.6 \times 10^{-8},\ 9.7 \times 10^{-8}]$	9.670234×10^{-8}	ok
muonic_thrust_to_power_shortfall	v_e(muonic)/v_e(Raptor)	$[5{,}800,\ 5{,}820]$	$5{,}810$	ok
domainBoundary_km_per_s	domain boundary, km/s	$[20{,}680,\ 20{,}700]$	$20{,}682$	ok
massRatioMuon_pow_five	207^5	exact numeral	380059617807	ok
massRatioTauon_pow_five	3477^5	exact numeral	5081868856832811577	ok
muonAmplification_eq	207^3	exact numeral	8869743	ok
tauonAmplification_eq	3477^3	exact numeral	42035292333	ok
collisionVelocityMuonMax_eq	25 000 km/h in m/s	exact numeral	62500 / 9	ok

5. Provenance. Source digests (SHA-256, first 16 hex digits) at report time:

Module	Digest
Basic	b15febfb52f101a4
Leptons	9ac624893fc8a88b
Dimensions	490237f0691b40ab
Velocity	d1716d3817282452
Regimes	0ce9852307164c64
TheoreticalFramework	bc1e1b637f57f99e

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A Declaration index

Every declaration of the development, with its docstring. The list is produced by parsing the sources, so no declaration can be omitted by an oversight: if it compiles, it is here.

Name	Kind	Content
`le_of_sq_le_sq`	`theorem`	Comparison of two non-negative reals through their squares. This is the tool that turns an exactly computed `vf²` into a bound on `vf`.
`effRadius`	`def`	Effective radius of a lepton, `r_ℓ = κ / m_ℓ`. `κ` has dimensions of mass × length. The framework fixes `κ` by requiring that the electron's effective radius reproduces its classical radius, `r_e = 2.8 × 10⁻¹⁵ m`; no numerical value is needed for the scaling laws, which are `κ`-free.
`effRadius_pos`	`theorem`	The effective radius of a positive-mass lepton is positive.
`effRadius_div`	`theorem`	Effective radii are inversely proportional to mass: the *compaction* property that drives the whole framework.
`effRadius_strictAnti`	`theorem`	A heavier lepton is the more compact one.
`casimirEnergy`	`def`	Casimir energy of the internal two-plate geometry, in units where the prefactor `π²ħc/720` is absorbed into the area coefficient. `E_C ∝ --- S / d³` with `S` the plate area and `d = r_ℓ = κ/m` the plate separation. The parallel-plate area ansatz `S ∝ m²` is the source's, and is recorded here explicitly by the `m^2` numerator.
`casimirEnergy_eq_quintic`	`theorem`	**Quintic law.** Under the inverse-radius ansatz the Casimir energy of a lepton is proportional to the fifth power of its mass.
`casimirEnergy_ratio`	`theorem`	**Amplification ratio.** Between two leptons, the Casimir amplification is the fifth power of their mass ratio. The constant `κ` cancels.
`casimirEnergy_ratio_radii`	`theorem`	The same amplification, read geometrically: it is the fifth power of the *inverse* ratio of effective radii. Compactness and vacuum coupling are two faces of one quantity.
`CasimirAnsatz`	`structure`	The geometric ansatz of the framework: an internal plate geometry whose separation is the effective radius and whose area scales as the square of the mass.
`CasimirAnsatz.plateArea`	`def`	Plate area of the internal geometry, `S = areaCoeff · m²`.
`CasimirAnsatz.energy`	`def`	Casimir energy of the ansatz, `E = -S/d³` up to a positive prefactor; its magnitude is `S/d³`.
`CasimirAnsatz.signedEnergy`	`def`	The signed (physical, attractive) Casimir energy, `E = -C · S/d³` with `C > 0`.
`CasimirAnsatz.energy_eq_quintic`	`theorem`	
`CasimirAnsatz.energy_ratio`	`theorem`	The amplification ratio of the ansatz is the fifth power of the mass ratio.

Name	Kind	Content
`CasimirAnsatz.energy_ratio_parameter_free`	`theorem`	**Parameter-freedom of the prediction.** Two arbitrary choices of the framework's free constants — κ, the area coefficient, even the overall Casimir prefactor — give the same amplification ratio. The ratio $(m_1/m_2)^5$ is thus a genuine prediction of the ansatz rather than a fit.
`CasimirAnsatz.signedEnergy_ratio`	`theorem`	The sign convention of the Casimir effect does not affect the amplification: the physical energy is negative (attractive), and the ratio is unchanged.
`massElectronMeV`	`def`	Electron mass, MeV/c².
`massMuonMeV`	`def`	Muon mass, MeV/c².
`massTauonMeV`	`def`	Tauon mass, MeV/c².
`massRatioMuon`	`def`	Muon-to-electron mass ratio, as used throughout the framework.
`massRatioTauon`	`def`	Tauon-to-electron mass ratio, as used throughout the framework.
`massRatioMuon_pos`	`theorem`	
`massRatioTauon_pos`	`theorem`	
`massRatioMuon_consistent`	`theorem`	The integer ratio 207 reproduces the tabulated muon/electron mass ratio to better than a quarter of a unit.
`massRatioTauon_consistent`	`theorem`	The integer ratio 3477 reproduces the tabulated tauon/electron mass ratio to better than half a unit.
`casimir_amplification_muon_general`	`theorem`	**Muonic Casimir amplification**, in the form `E_μ / E_e = (m_μ/m_e)^5`.
`casimir_amplification_tauon_general`	`theorem`	**Tauonic Casimir amplification**, in the form `E_τ / E_e = (m_τ/m_e)^5`.
`massRatioMuon_pow_five`	`theorem`	Exact value of the muonic amplification, $207^5 = 380\,059\,617\,807$.
`massRatioTauon_pow_five`	`theorem`	Exact value of the tauonic amplification, $3477^5 = 508\,186\,885\,683\,281\,157$.
`casimir_amplification_muon_order`	`theorem`	$(207)^5 \approx 4 \times 10^{11}$: the amplification lies between 3.8×10^{11} and 4×10^{11}.
`casimir_amplification_tauon_order`	`theorem`	$(3477)^5 \approx 5.1 \times 10^{17}$: the amplification lies between 5.08×10^{17} and 5.1×10^{17}.
`casimir_amplification_tauon_over_muon`	`theorem`	The tauon amplifies the vacuum coupling by a further factor $(3477/207)^5 \approx 1.3 \times 10^6$ over the muon.
`casimir_amplification_hierarchy`	`theorem`	The amplification is monotone in the mass ratio: the lepton hierarchy $e < \mu < \tau$ is a strict hierarchy of vacuum couplings.
`fieldAmplificationSource`	`def`	Equivalent-field amplification under the source's volume ansatz $V \propto S \cdot d^3$: $B \propto m^3$.
`fieldAmplificationPlate`	`def`	Equivalent-field amplification under the literal plate volume $V = S \cdot d$: $B \propto m^2$.
`field_amplification_source`	`theorem`	Source ansatz: $B^2 \propto E_C / (S \cdot d^3) \propto m^6$, hence $B \propto m^3$.
`field_amplification_plate`	`theorem`	Literal plate volume: $B^2 \propto E_C / (S \cdot d) \propto m^4$, hence $B \propto m^2$.
`field_amplification_muon_order`	`theorem`	The source's field amplification for the muon, $(207)^3 \approx 9 \times 10^6$.
`casimirMassExponent`	`def`	Exponent of the mass in the Casimir energy of the compactness ansatz.
`generalizedDimension`	`def`	The generalized dimension, fixed by the quintic mass dependence: $n = 5$.

Name	Kind	Content
generalizedDimension_eq_five	theorem	The generalized dimension is five.
structuralTerm	def	The structural part of the anomalous moment, a_str = C · m^n.
anomalousMoment	def	The framework's decomposition a_ℓ = a_QED + a_EW+had + a_str.
structuralTerm_quintic	theorem	The structural term is quintic in the mass.
structuralTerm_ratio	theorem	The structural term amplifies between two leptons by the fifth power of the mass ratio: the same number as the Casimir amplification, which is the reason a single exponent serves both.
velocityExponent	def	Exponent of the mass ratio in the velocity law, n − 2.
velocityExponent_eq_three	theorem	With n = 5 the velocity law carries the cube of the mass ratio.
velocityExponent_eq	theorem	The velocity exponent is n − 2 for the framework's n.
generalised_unit_exponents	theorem	Unit bookkeeping of [v_n] = m^(2/n)/s: the mass exponent balances to 2.
time_exponent_balance	theorem	Unit bookkeeping of the time factor dt_0^(n−2): the exponents close on n.
GeneralisedVelocityRelation	def	The generalized-velocity relation $v_n{}^n$ · dt_0^(n−2) = $v_2{}^2$ of the framework.
scale_substitution	theorem	**Scale substitution.** Replacing the time scale by q·dt_0 multiplies the relation by q^(n−2), which is the framework's substitution rule v_n·(q·dt_0)^(n−2) = $v_2{}^2$·q^(n−2); the parameter q = m_ℓ/m_0 is what links the muonic and tauonic regimes.
generalisedVelocity	def	Ejection velocity of particles produced in lepton–lepton collisions in the generalized framework: the non-negative root of vf^2 = $v_2{}^2$·r^(n−2), where r is the mass ratio m_ℓ/m_0.
velocity_law	theorem	**The velocity law**, for arbitrary generalized dimension n: vf^2 = $v_2{}^2$ · (m_ℓ/m_0)^(n−2).
generalisedVelocity_eq_mul	theorem	The paper's form of the law, vf = v_2 · √((m_ℓ/m_0)^(n−2)), agrees with the definition for non-negative input velocity.
generalisedVelocity_nonneg	theorem	Ejection velocities are non-negative.
generalisedVelocity_bounds	theorem	Two-sided bounds on an ejection velocity, from exact bounds on vf^2. This is the workhorse that turns a computed square into a velocity interval without any rounding step.
velocity_law_five	theorem	The velocity law at the framework's generalized dimension: vf^2 = $v_2{}^2$ · (m_ℓ/m_0)^3.
generalisedVelocity_five_eq_mul	theorem	The law's explicit linear-in-v_2 form at n = 5: vf = v_2·√(r^3).
generalisedVelocity_mono	theorem	The ejection velocity grows with the lepton mass ratio: heavier collision partners eject faster.
generalisedVelocity_mono_v_2	theorem	The ejection velocity grows with the collision velocity.
generalisedVelocity_zero	theorem	The ejection velocity vanishes with the collision velocity: no collisions, no propulsion.

Name	Kind	Content
kmhToMs	def	Kilometres per hour to metres per second, 1 km/h = 5/18 m/s.
kmhToMs_eq	theorem	The conversion agrees with the definitional 1000/3600.
standardGravity	def	Standard gravity, g_0 = 9.80665 m/s², used to express specific impulse.
standardGravity_pos	theorem	
speedOfLight	def	Speed of light in vacuum, m/s.
specificImpulse	def	Specific impulse, Isp = v_e / g_0, in seconds.
raptorSpecificImpulse	def	Vacuum specific impulse of a Raptor engine, seconds: the chemical benchmark of the source text.
collisionVelocityMuonMax	def	Upper end of the muonic input window, v_2 = 25 000 km/h.
collisionVelocityMuonMax_eq	theorem	The muonic input ceiling in m/s.
collisionVelocityMuonMax_bounds	theorem	25 000 km/h ≈ 6 944 m/s.
muonAmplification	def	Muonic amplification factor, (m_μ/m_e)^(n−2).
tauonAmplification	def	Tauonic amplification factor, (m_τ/m_e)^(n−2).
muonAmplification_eq	theorem	207^3 = 8 869 743.
tauonAmplification_eq	theorem	3477^3 = 42 035 292 333.
muonAmplification_order	theorem	The muonic amplification is $(207)^3 \approx 8.9 \times 10^6$.
tauonAmplification_order	theorem	The tauonic amplification is $(3477)^3 \approx 4.2 \times 10^{10}$.
tauon_over_muon_amplification	theorem	The tauonic amplification exceeds the muonic one by $(3477/207)^3 \approx 4.74 \times 10^3$.
muonMaxEjection	def	Maximum ejection velocity of the muonic regime, at v_2 = 25 000 km/h.
muonMaxEjection_sq	theorem	The ejection velocity obeys the law: $vf^2 = {v_2}^2 \cdot 207^3$.
muonMaxEjection_bounds	theorem	vf ≈ 2.06820 × 10^7 m/s for the muonic ceiling.
muonMaxEjection_km_per_s	theorem	The source's ≈ 20 700 km/s, proved as an interval in km/s.
muonMaxEjection_subluminal	theorem	The muonic ejection velocity is subluminal: the muonic regime is compatible with ordinary special relativity.
collisionVelocityMuonMax_subluminal	theorem	The muonic input window is subluminal as well.
muonSpecificImpulse_bounds	theorem	Specific impulse of the muonic ceiling, Isp = $v_e/g_0 \approx 2.109 \times 10^6$ s.
muonImpulse_over_raptor	theorem	The muonic impulse exceeds the Raptor benchmark by a factor of about 5.8×10^3, i.e. "roughly 6 000 times" as the source states.
tauonicInputMax	def	Input ceiling of the tauonic regime: the muonic regime's maximum ejection velocity, ≈ 20 682 km/s. Taking the unrounded value is what makes the two regimes exactly continuous.
tauonMaxEjection	def	Maximum ejection velocity of the tauonic regime, at v_2 = tauonicInputMax.
tauonMaxEjection_sq	theorem	The tauonic ejection velocity obeys the law: $vf^2 = {v_2}^2 \cdot 3477^3$.
tauonMaxEjection_bounds	theorem	vf ≈ 4.2403 × 10^{12} m/s for the tauonic ceiling.

Name	Kind	Content
`tauonMaxEjection_superluminal`	`theorem`	The tauonic ceiling is superluminal: the framework's rejection of the Lorentz factor is not optional decoration, it is what its own numbers require.
`tauonicInputMax_subluminal`	`theorem`	The tauonic input window is subluminal; only the output exceeds `c`.
`sourceTauonMaxVelocity`	`def`	The tauonic ceiling ejection velocity as printed in the source text.
`sourceTauonMedianVelocity`	`def`	The tauonic median ejection velocity as printed in the source text.
`source_tauon_max_velocity_underestimates`	`theorem`	The source figure, squared, falls short of the law by a factor of `10.01`: the printed value is √10 too small.
`source_tauon_max_velocity_factor`	`theorem`	Consequently the source figure underestimates the ejection velocity itself by the factor √10 ≈ 3.164.
`tauonEntryEjection`	`def`	Tauonic ejection velocity at the entry velocity of the tauonic window.
`tauonEntryEjection_bounds`	`theorem`	`vf` ≈ 1.4238 × 10^9 m/s at the tauonic entry velocity: the source's 1.42 × 10^9 is correct here, because this `vf`² happens to carry an even decimal exponent.
`tauonMedianEjection`	`def`	Tauonic ejection velocity at a median input of 500 000 km/h.
`tauonMedianEjection_bounds`	`theorem`	`vf` ≈ 2.8477 × 10^{10} m/s at the median input — not the 9.0 × 10^{10} printed in the source, which carries the same √10 slip.
`amplification_mul`	`theorem`	The amplification factor is multiplicative in the mass ratio: chaining two mass scales multiplies the amplification. This is what makes the regimes composable rather than merely analogous.
`amplification_strictMono`	`theorem`	The amplification is strictly increasing in the mass ratio on the positive axis: heavier collision partners amplify more.
`tauon_amplification_factors`	`theorem`	The tauonic amplification is the muonic one times the amplification of the residual ratio m_τ/m_μ: $3477^3 = 207^3 \cdot (3477/207)^3$.
`generalisedVelocity_ge_input`	`theorem`	For a mass ratio at least one, the ejection velocity is at least the collision velocity. This is what lets the tauonic window start where the muonic window ends.
`one_le_massRatioMuon`	`theorem`	
`one_le_massRatioTauon`	`theorem`	
`PropulsiveRegime`	`structure`	A propulsive regime of the framework: a charged-lepton mass ratio, the window of admissible collision velocities, and the law-induced quantities derived from them.
`PropulsiveRegime.amplification`	`def`	Amplification factor of the regime, `(m_ℓ/m_e)^(n−2)`.
`PropulsiveRegime.maxOutput`	`def`	Maximum ejection velocity of the regime, at the top of its input window.
`PropulsiveRegime.window`	`def`	The input window as a set.
`PropulsiveRegime.maxOutput_sq`	`theorem`	The regime's maximum output obeys the velocity law.
`PropulsiveRegime.inputUpper_le_maxOutput`	`theorem`	The maximum output is at least the input ceiling.

Name	Kind	Content
`PropulsiveRegime.continuous_maxOutput`	`theorem`	Within a regime the velocity law is continuous in the collision velocity.
`PropulsiveRegime.continuous_in_massRatio`	`theorem`	Within a regime the velocity law is continuous in the mass ratio.
`muonicRegime`	`def`	**The muonic regime.** Muon–muon collisions, mass ratio 207, collision velocities up to 25 000 km/h; replaces chemical propellants.
`tauonicRegime`	`def`	**The tauonic regime.** Tauon–tauon collisions, mass ratio 3477, input window from the muonic ceiling up to the muonic regime's output ceiling.
`muonicRegime_amplification`	`theorem`	The muonic amplification, through the structure.
`tauonicRegime_amplification`	`theorem`	The tauonic amplification, through the structure.
`regimes_strictly_ordered`	`theorem`	The regime hierarchy is strict: the tauonic amplification exceeds the muonic one by $\approx 4.74 \times 10^3$.
`muonicRegime_maxOutput`	`theorem`	The muonic regime's output ceiling, through the structure.
`tauonicRegime_maxOutput`	`theorem`	The tauonic regime's output ceiling, through the structure.
`windows_adjacent`	`theorem`	**Adjacency.** The tauonic window begins exactly where the muonic window ends: 25 000 km/h, the muonic collision ceiling.
`chain_closes`	`theorem`	**The chain closes.** The upper end of the tauonic *input* window is the muonic regime's maximum *output*. This is the boundary the framework calls v_2 = 20 700 km/s, and taking the unrounded value is what makes the match exact rather than approximate.
`regimes_tile`	`theorem`	**Tiling.** The two input windows cover one interval with no gap and no overlap.
`tauonic_window_nested_in_output`	`theorem`	**Nesting.** The tauonic window lies inside the muonic regime's output range: the second regime's inputs are the first regime's reachable outputs.
`domainBoundary`	`def`	**The domain boundary.** The boundary of the source's §8.2, v_2 = 20 700 km/s, is the tauonic input ceiling, and it is the muonic output ceiling unrounded.
`domainBoundary_eq_muonic_output`	`theorem`	The boundary is the muonic regime's maximum output.
`domainBoundary_km_per_s`	`theorem`	The boundary is the muonic ejection ceiling, hence ≈ 20 700 km/s.
`domainBoundary_subluminal`	`theorem`	The boundary itself is subluminal; it is the tauonic *output* that exceeds c, not the input ceiling that defines the hand-off.
`regime_continuity`	`theorem`	**Continuity of the two-regime structure**, collected in one statement: the windows are adjacent, the chain closes on the muonic output ceiling, they tile an interval, and the law is continuous in the collision velocity throughout.
`ThrusterDesign`	`structure`	A thruster design obeying the framework's momentum and energy balance.
`raptorExhaustVelocity`	`def`	Vacuum exhaust velocity of the Raptor benchmark.

Name	Kind	Content
thrust_to_power	theorem	**The thrust–power tradeoff.** For any design, T/P = 2/v_e exactly. High specific impulse costs thrust per watt, without exception.
thrust_eq_power	theorem	Thrust in terms of jet power and exhaust velocity.
muonic_thrust_to_power_bounds	theorem	At the muonic ceiling, thrust per watt is ≈ 9.67 × 10⁻⁸ N/W.
muonic_thrust_to_power_shortfall	theorem	The muonic exhaust velocity is ≈ 5.8 × 10³ times the Raptor exhaust velocity, so its thrust per watt is that many times *worse*: the specific-impulse gain and the thrust-per-watt loss are the same number.
thrust_not_determined	theorem	**Thrust is not determined by the geometry.** At the exhaust velocity the leptonic kinematics fix, any positive thrust is attainable by a suitable design: thrust is bought with mass flow and power, not with the velocity law.
ThrustOptimizationConjecture	def	The engineering question the framework leaves open: at the exhaust velocity the geometry fixes, is there a design reaching a thrust target T within a power budget P? Nothing in the geometric development decides it — see thrust_not_determined — so it is recorded, not proved.
thrust_optimization_satisfiable	theorem	The conjecture is not vacuous: a large enough power budget always suffices, because the thrust–power relation is a hyperbola and nothing else constrains it.
thrust_optimization_is_empirical	theorem	Consequently the thrust question is *empirical*, not geometric: the framework fixes v_e and hence T/P, and leaves (T, P) free along that hyperbola. This is the formal statement of ”empirical thrust optimisations remain unproven”.

B The verified sources

The complete development, as elaborated by Lean. Digests are in §8.

TheoreticalFramework/Basic.lean

```
/-
# TheoreticalFramework.Basic

Core derivations of the leptonic-compactness framework.

The model under study attributes part of the anomalous magnetic moment of a
massive lepton to an *internal geometric compactness*: a lepton of mass `m`
carries an effective radius

`r_ℓ = κ / m`,

with `κ` a fixed constant of the framework (dimensions of mass × length).  The
internal geometry is modelled, following the source text, as a pair of plates
whose separation is `d = r_ℓ` and whose area scales with the square of the mass,
`S ∝ m²`.  The Casimir energy of such a cavity scales as

`E_C ∝ - S / d³`,

hence as `- m² / r_ℓ³ = - m⁵ / κ³`.

The mathematical content proved here is the scaling: the Casimir energy of a
lepton is quintic in its mass, and therefore the amplification ratio between two
leptons depends only on the ratio of their masses — not on `κ`, not on the area
coefficient, and not on the overall Casimir constant.  That parameter-freedom is
what lets the framework issue a numerical prediction, and it is the reason the
derivation can be checked exactly rather than up to an unknown prefactor.

The geometric *ansatz* itself (`S ∝ m²`, `d = r_ℓ`) is recorded as an explicit
model assumption in `CasimirAnsatz`, not asserted as a derived fact.
-/
import Mathlib

namespace TheoreticalFramework

noncomputable section

/-- Comparison of two non-negative reals through their squares.  This is the
tool that turns an exactly computed `vf²` into a bound on `vf`. -/
theorem le_of_sq_le_sq {a b : ℝ} (ha : 0 ≤ a) (hb : 0 ≤ b) (h : a ^ 2 ≤ b ^ 2) : a ≤ b := by
have habs := sq_le_sq.mp h
rwa [abs_of_nonneg ha, abs_of_nonneg hb] at habs

/-! ## Effective radius -/

/-- Effective radius of a lepton, `r_ℓ = κ / m_ℓ`.

`κ` has dimensions of mass × length.  The framework fixes `κ` by requiring that
the electron's effective radius reproduces its classical radius, `r_e = 2.8 × 10⁻¹⁵ m`;
no numerical value is needed for the scaling laws, which are `κ`-free. -/
def effRadius (κ m : ℝ) : ℝ := κ / m

/-- The effective radius of a positive-mass lepton is positive. -/
theorem effRadius_pos {κ m : ℝ} (hκ : 0 < κ) (hm : 0 < m) : 0 < effRadius κ m :=
div_pos hκ hm

/-- Effective radii are inversely proportional to mass: the *compaction* property
that drives the whole framework. -/
theorem effRadius_div {κ m₁ m₂ : ℝ} (hκ : κ ≠ 0) (h₁ : m₁ ≠ 0) (h₂ : m₂ ≠ 0) :
effRadius κ m₁ / effRadius κ m₂ = m₂ / m₁ := by
unfold effRadius
field_simp

/-- A heavier lepton is the more compact one. -/
theorem effRadius_strictAnti {κ : ℝ} (hκ : 0 < κ) :
StrictAntiOn (effRadius κ) (Set.Ioi 0) := by
intro m₁ hm₁ m₂ hm₂ hlt
simp only [Set.mem_Ioi] at hm₁ hm₂
exact div_lt_div_of_pos_left hκ hm₁ hlt

/-! ## Casimir energy of the internal geometry -/

/-- Casimir energy of the internal two-plate geometry, in units where the
prefactor `π²ħc/720` is absorbed into the area coefficient.

`E_C ∝ - S / d³` with `S` the plate area and `d = r_ℓ = κ/m` the plate
separation.  The parallel-plate area ansatz `S ∝ m²` is the source's, and is
recorded here explicitly by the `m^2` numerator. -/
def casimirEnergy (κ m : ℝ) : ℝ := m ^ 2 / (effRadius κ m) ^ 3

/-- **Quintic law.**  Under the inverse-radius ansatz the Casimir energy of a
lepton is proportional to the fifth power of its mass. -/
```

```
theorem casimirEnergy_eq_quintic {κ m : ℝ} (hκ : κ ≠ 0) (hm : m ≠ 0) :
casimirEnergy κ m = m ^ 5 / κ ^ 3 := by
unfold casimirEnergy effRadius
field_simp

/-- **Amplification ratio.**  Between two leptons, the Casimir amplification is
the fifth power of their mass ratio.  The constant `κ` cancels. -/
theorem casimirEnergy_ratio {κ : ℝ} (hκ : κ ≠ 0) {m₁ m₂ : ℝ} (h₁ : m₁ ≠ 0) (h₂ : m₂ ≠ 0) :
casimirEnergy κ m₁ / casimirEnergy κ m₂ = (m₁ / m₂) ^ 5 := by
rw [casimirEnergy_eq_quintic hκ h₁, casimirEnergy_eq_quintic hκ h₂]
field_simp

/-- The same amplification, read geometrically: it is the fifth power of the
*inverse* ratio of effective radii.  Compactness and vacuum coupling are two
faces of one quantity. -/
theorem casimirEnergy_ratio_radii {κ m₁ m₂ : ℝ} (hκ : κ ≠ 0) (h₁ : m₁ ≠ 0) (h₂ : m₂ ≠ 0) :
casimirEnergy κ m₁ / casimirEnergy κ m₂ = (effRadius κ m₂ / effRadius κ m₁) ^ 5 := by
rw [casimirEnergy_ratio hκ h₁ h₂, effRadius_div hκ h₂ h₁]

/-! ## The geometric ansatz as an explicit structure

Isolating the ansatz makes two things visible: which quantities are free, and
that the amplification ratio does not depend on any of them.
-/

/-- The geometric ansatz of the framework: an internal plate geometry whose
separation is the effective radius and whose area scales as the square of the
mass. -/
structure CasimirAnsatz where
/-- Constant of the inverse-radius law, `r_ℓ = κ/m_ℓ`. -/
kappa : ℝ
kappa_pos : 0 < kappa
/-- Coefficient of the plate-area ansatz, `S = areaCoeff · m²`. -/
areaCoeff : ℝ
areaCoeff_pos : 0 < areaCoeff

namespace CasimirAnsatz

/-- Plate area of the internal geometry, `S = areaCoeff · m²`. -/
def plateArea (A : CasimirAnsatz) (m : ℝ) : ℝ := A.areaCoeff * m ^ 2

/-- Casimir energy of the ansatz, `E = -S/d³` up to a positive prefactor;
its magnitude is `S/d³`. -/
def energy (A : CasimirAnsatz) (m : ℝ) : ℝ := A.plateArea m / (effRadius A.kappa m) ^ 3

/-- The signed (physical, attractive) Casimir energy, `E = -C · S/d³` with `C > 0`. -/
def signedEnergy (A : CasimirAnsatz) (C m : ℝ) : ℝ := -C * A.energy m

theorem energy_eq_quintic (A : CasimirAnsatz) {m : ℝ} (hm : m ≠ 0) :
A.energy m = A.areaCoeff * m ^ 5 / A.kappa ^ 3 := by
unfold energy plateArea effRadius
field_simp [A.kappa_pos.ne', A.areaCoeff_pos.ne',
mul_ne_zero A.areaCoeff_pos.ne' A.kappa_pos.ne']

/-- The amplification ratio of the ansatz is the fifth power of the mass ratio. -/
theorem energy_ratio (A : CasimirAnsatz) {m₁ m₂ : ℝ} (h₁ : m₁ ≠ 0) (h₂ : m₂ ≠ 0) :
A.energy m₁ / A.energy m₂ = (m₁ / m₂) ^ 5 := by
rw [energy_eq_quintic A h₁, energy_eq_quintic A h₂]
field_simp [A.kappa_pos.ne', A.areaCoeff_pos.ne', pow_ne_zero 5 h₁, pow_ne_zero 5 h₂]

/-- **Parameter-freedom of the prediction.**  Two arbitrary choices of the
framework's free constants — `κ`, the area coefficient, even the overall Casimir
prefactor — give the same amplification ratio.  The ratio `(m₁/m₂)⁵` is thus a
genuine prediction of the ansatz rather than a fit. -/
theorem energy_ratio_parameter_free (A A' : CasimirAnsatz) {m₁ m₂ : ℝ}
(h₁ : m₁ ≠ 0) (h₂ : m₂ ≠ 0) :
A.energy m₁ / A.energy m₂ = A'.energy m₁ / A'.energy m₂ := by
rw [energy_ratio A h₁ h₂, energy_ratio A' h₁ h₂]

/-- The sign convention of the Casimir effect does not affect the amplification:
the physical energy is negative (attractive), and the ratio is unchanged. -/
theorem signedEnergy_ratio (A : CasimirAnsatz) {C m₁ m₂ : ℝ} (hC : C ≠ 0)
(h₁ : m₁ ≠ 0) (h₂ : m₂ ≠ 0) :
A.signedEnergy C m₁ / A.signedEnergy C m₂ = (m₁ / m₂) ^ 5 := by
unfold signedEnergy
rw [neg_mul, neg_mul, neg_div_neg_eq, mul_div_mul_left _ _ hC, energy_ratio A h₁ h₂]

end CasimirAnsatz

end

end TheoreticalFramework
```

TheoreticalFramework/Leptons.lean

```
/-
# TheoreticalFramework.Leptons

Numerical instantiation of the compactness framework for the three charged
leptons, and the Casimir amplification ratios of the muonic and tauonic regimes.

The lepton masses enter only through the ratios `m_μ/m_e ≈ 207` and
`m_τ/m_e ≈ 3477`.  Both integers are checked here against the tabulated masses
(`0.511`, `105.66`, `1777` MeV/c²) at the precision at which the framework
states them, and the amplification ratios are then computed exactly and bounded
in the published orders of magnitude.
-/
import TheoreticalFramework.Basic

namespace TheoreticalFramework

noncomputable section

/-! ## Lepton data -/

/-- Electron mass, MeV/c². -/
def massElectronMeV : ℝ := 0.511

/-- Muon mass, MeV/c². -/
def massMuonMeV : ℝ := 105.66

/-- Tauon mass, MeV/c². -/
def massTauonMeV : ℝ := 1777

/-- Muon-to-electron mass ratio, as used throughout the framework. -/
def massRatioMuon : ℝ := 207

/-- Tauon-to-electron mass ratio, as used throughout the framework. -/
def massRatioTauon : ℝ := 3477

theorem massRatioMuon_pos : 0 < massRatioMuon := by
norm_num [massRatioMuon]

theorem massRatioTauon_pos : 0 < massRatioTauon := by
norm_num [massRatioTauon]

/-- The integer ratio `207` reproduces the tabulated muon/electron mass ratio to
better than a quarter of a unit. -/
theorem massRatioMuon_consistent :
|massMuonMeV / massElectronMeV - massRatioMuon| ≤ 0.25 := by
unfold massMuonMeV massElectronMeV massRatioMuon
rw [abs_le]
constructor <;> norm_num

/-- The integer ratio `3477` reproduces the tabulated tauon/electron mass ratio
to better than half a unit. -/
theorem massRatioTauon_consistent :
|massTauonMeV / massElectronMeV - massRatioTauon| ≤ 0.5 := by
unfold massTauonMeV massElectronMeV massRatioTauon
rw [abs_le]
constructor <;> norm_num

/-! ## Casimir amplification, muon and tauon versus electron

The general ratio `casimirEnergy_ratio` is instantiated with masses
`m_ℓ = (m_ℓ/m_e) · m_e`, so the result is a statement about the physics, not
about a bare numeral.  The numeral is then evaluated exactly.
-/

/-- **Muonic Casimir amplification**, in the form `E_μ / E_e = (m_μ/m_e)⁵`. -/
theorem casimir_amplification_muon_general {κ m₀ : ℝ} (hκ : κ ≠ 0) (hm : 0 < m₀) :
casimirEnergy κ (massRatioMuon * m₀) / casimirEnergy κ m₀ = massRatioMuon ^ 5 := by
rw [casimirEnergy_ratio hκ (mul_ne_zero massRatioMuon_pos.ne' hm.ne') hm.ne']
field_simp

/-- **Tauonic Casimir amplification**, in the form `E_τ / E_e = (m_τ/m_e)⁵`. -/
theorem casimir_amplification_tauon_general {κ m₀ : ℝ} (hκ : κ ≠ 0) (hm : 0 < m₀) :
casimirEnergy κ (massRatioTauon * m₀) / casimirEnergy κ m₀ = massRatioTauon ^ 5 := by
rw [casimirEnergy_ratio hκ (mul_ne_zero massRatioTauon_pos.ne' hm.ne') hm.ne']
field_simp

/-- Exact value of the muonic amplification, `207⁵ = 380 059 617 807`. -/
theorem massRatioMuon_pow_five : massRatioMuon ^ 5 = 380059617807 := by
norm_num [massRatioMuon]

/-- Exact value of the tauonic amplification, `3477⁵ = 508 186 885 683 281 157`. -/
theorem massRatioTauon_pow_five : massRatioTauon ^ 5 = 508186885683281157 := by
norm_num [massRatioTauon]

/-- `(207)⁵ ≈ 4 × 10¹¹`: the amplification lies between `3.8 × 10¹¹` and `4 × 10¹¹`. -/
theorem casimir_amplification_muon_order :
(3.8 : ℝ) * 10 ^ 11 ≤ massRatioMuon ^ 5 ∧ massRatioMuon ^ 5 ≤ 4 * 10 ^ 11 := by
rw [massRatioMuon_pow_five]
constructor <;> norm_num

/-- `(3477)⁵ ≈ 5.1 × 10¹⁷`: the amplification lies between `5.08 × 10¹⁷` and `5.1 × 10¹⁷`. -/
```

```
theorem casimir_amplification_tauon_order :
(5.08 : ℝ) * 10 ^ 17 ≤ massRatioTauon ^ 5 ∧ massRatioTauon ^ 5 ≤ (5.1 : ℝ) * 10 ^ 17 := by
rw [massRatioTauon_pow_five]
constructor <;> norm_num

/-- The tauon amplifies the vacuum coupling by a further factor
`(3477/207)⁵ ≈ 1.3 × 10⁶` over the muon. -/
theorem casimir_amplification_tauon_over_muon :
(1.3 : ℝ) * 10 ^ 6 ≤ (massRatioTauon / massRatioMuon) ^ 5 ∧
(massRatioTauon / massRatioMuon) ^ 5 ≤ (1.34 : ℝ) * 10 ^ 6 := by
unfold massRatioTauon massRatioMuon
constructor <;> norm_num

/-- The amplification is monotone in the mass ratio: the lepton hierarchy
`e < μ < τ` is a strict hierarchy of vacuum couplings. -/
theorem casimir_amplification_hierarchy :
(1 : ℝ) < massRatioMuon ^ 5 ∧ massRatioMuon ^ 5 < massRatioTauon ^ 5 := by
rw [massRatioMuon_pow_five, massRatioTauon_pow_five]
constructor <;> norm_num

/-! ## Equivalent magnetic field, and where the exponent `3` comes from

The source text associates the Casimir energy with an equivalent magnetic field
through `B² ∝ E_C / Volume`.  Two volume ansätze are possible, and they give
different exponents.  Both are proved here so that the framework's use of
`B/B₀ ∝ m³` is tied to a stated assumption rather than left implicit.

* With the source's volume ansatz `V ∝ S·d³` one gets `B² ∝ m⁶`, i.e. `B ∝ m³`.
* With the geometrically literal plate volume `V = S·d` one gets `B² ∝ m⁴`, i.e. `B ∝ m²`.

The first is the exponent the framework uses, and it coincides with the velocity
exponent `n - 2 = 3`; the second is recorded because it is what a literal plate
volume gives, and the discrepancy is a property of the ansatz, not of the
algebra.
-/

/-- Equivalent-field amplification under the source's volume ansatz `V ∝ S·d³`:
`B ∝ m³`. -/
def fieldAmplificationSource (m₁ m₂ : ℝ) : ℝ := (m₁ / m₂) ^ 3

/-- Equivalent-field amplification under the literal plate volume `V = S·d`:
`B ∝ m²`. -/
def fieldAmplificationPlate (m₁ m₂ : ℝ) : ℝ := (m₁ / m₂) ^ 2

/-- Source ansatz: `B² ∝ E_C / (S·d³) ∝ m⁶`, hence `B ∝ m³`. -/
theorem field_amplification_source (κ m₁ m₂ : ℝ) (hκ : 0 < κ) (h₁ : 0 < m₁) (h₂ : 0 < m₂) :
(casimirEnergy κ m₁ / (m₁ ^ 2 * effRadius κ m₁ ^ 3)) /
(casimirEnergy κ m₂ / (m₂ ^ 2 * effRadius κ m₂ ^ 3)) = (m₁ / m₂) ^ 6 := by
rw [casimirEnergy_eq_quintic hκ.ne' h₁.ne', casimirEnergy_eq_quintic hκ.ne' h₂.ne']
unfold effRadius
field_simp

/-- Literal plate volume: `B² ∝ E_C / (S·d) ∝ m⁴`, hence `B ∝ m²`. -/
theorem field_amplification_plate (κ m₁ m₂ : ℝ) (hκ : 0 < κ) (h₁ : 0 < m₁) (h₂ : 0 < m₂) :
(casimirEnergy κ m₁ / (m₁ ^ 2 * effRadius κ m₁)) /
(casimirEnergy κ m₂ / (m₂ ^ 2 * effRadius κ m₂)) = (m₁ / m₂) ^ 4 := by
rw [casimirEnergy_eq_quintic hκ.ne' h₁.ne', casimirEnergy_eq_quintic hκ.ne' h₂.ne']
unfold effRadius
field_simp

/-- The source's field amplification for the muon, `(207)³ ≈ 9 × 10⁶`. -/
theorem field_amplification_muon_order :
(8.8 : ℝ) * 10 ^ 6 ≤ fieldAmplificationSource massRatioMuon 1 ∧
fieldAmplificationSource massRatioMuon 1 ≤ (8.9 : ℝ) * 10 ^ 6 := by
unfold fieldAmplificationSource massRatioMuon
constructor <;> norm_num

end

end TheoreticalFramework
```

TheoreticalFramework/Dimensions.lean

```
/-
# TheoreticalFramework.Dimensions

The generalized `n`-dimensional kinematic framework.

The framework's generalized dimension is not a free parameter: it is read off the
quintic mass dependence of the anomalous-moment contribution,

`a_ℓ = a_QED + a_EW+had(m_ℓ⁵) + a_str(m_ℓ, G_ℓ)`,

whose structural term scales as `m⁵` (`casimirEnergy_eq_quintic`).  Setting
`n = 5` makes the velocity exponent `n − 2 = 3`, which is the exponent that
appears in the ejection-velocity law and, independently, in the equivalent
magnetic-field amplification of `TheoreticalFramework.Leptons`.

The velocity law itself is stated for arbitrary `n ≥ 2` and then specialised:

`vf² = v₂² · (m_ℓ/m₀)^(n−2)`.

`vf` is defined as the non-negative root of that equation, and the paper's
multiplicative form `vf = v₂ · √((m_ℓ/m₀)^(n−2))` is proved equivalent for
`v₂ ≥ 0`.

The generalized units of the framework, `[Cₙ] = [vₙ] = m^(2/n)/s`, enter only
through exponent bookkeeping; that bookkeeping is proved in
`generalised_unit_exponents` and `time_exponent_balance`.
-/
import TheoreticalFramework.Leptons

namespace TheoreticalFramework

noncomputable section

/-! ## The generalized dimension -/

/-- Exponent of the mass in the Casimir energy of the compactness ansatz. -/
def casimirMassExponent : ℕ := 5

/-- The generalized dimension, fixed by the quintic mass dependence: `n = 5`. -/
def generalizedDimension : ℕ := casimirMassExponent

/-- The generalized dimension is five. -/
theorem generalizedDimension_eq_five : generalizedDimension = 5 := rfl

/-! ## The anomalous moment whose quintic term fixes `n`

The framework decomposes the anomalous moment of a lepton into a QED part, an
electroweak-plus-hadronic part and a structural part, the last being quintic in
the mass.  The quintic exponent is what fixes `n = 5`; it is not an independent
input. -/

/-- The structural part of the anomalous moment, `a_str = C · m^n`. -/
def structuralTerm (C m : ℝ) : ℝ := C * m ^ generalizedDimension

/-- The framework's decomposition `a_ℓ = a_QED + a_EW+had + a_str`. -/
def anomalousMoment (aQED aEWHad C m : ℝ) : ℝ := aQED + aEWHad + structuralTerm C m

/-- The structural term is quintic in the mass. -/
theorem structuralTerm_quintic (C m : ℝ) : structuralTerm C m = C * (m : ℝ) ^ 5 := by
unfold structuralTerm
rw [generalizedDimension_eq_five]

/-- The structural term amplifies between two leptons by the fifth power of the
mass ratio: the same number as the Casimir amplification, which is the reason a
single exponent serves both. -/
theorem structuralTerm_ratio (C m₁ m₂ : ℝ) (hC : C ≠ 0) (h₁ : m₁ ≠ 0) (h₂ : m₂ ≠ 0) :
structuralTerm C m₁ / structuralTerm C m₂ = (m₁ / m₂) ^ 5 := by
unfold structuralTerm
rw [generalizedDimension_eq_five, div_pow]
field_simp [hC, pow_ne_zero 5 h₁, pow_ne_zero 5 h₂]

/-- Exponent of the mass ratio in the velocity law, `n − 2`. -/
def velocityExponent : ℕ := generalizedDimension - 2

/-- With `n = 5` the velocity law carries the cube of the mass ratio. -/
theorem velocityExponent_eq_three : velocityExponent = 3 := rfl

/-- The velocity exponent is `n − 2` for the framework's `n`. -/
theorem velocityExponent_eq (n : ℕ) (hn : n = generalizedDimension) :
velocityExponent = n - 2 := by
subst hn
rfl

/-! ## Generalized units

The framework assigns to the generalized velocity the units `m^(2/n)/s`, so that
the `n`-th power carries `m²/sⁿ`; multiplying by `dt₀^(n−2)` then carries
`m²/s²`, the units of `v₂²`.  Only the exponents matter, and they are exact.
-/
```

```
/-- Unit bookkeeping of `[vₙ] = m^(2/n)/s`: the mass exponent balances to `2`. -/
theorem generalised_unit_exponents (n : ℚ) (hn : n ≠ 0) : n * (2 / n) = 2 := by
field_simp

/-- Unit bookkeeping of the time factor `dt₀^(n−2)`: the exponents close on `n`. -/
theorem time_exponent_balance (n : ℚ) : (n - 2) + 2 = n := by
ring

/-- The generalized-velocity relation `vₙⁿ · dt₀^(n−2) = v₂²` of the framework. -/
def GeneralisedVelocityRelation (n : ℕ) (vn dt0 v2 : ℝ) : Prop :=
vn ^ n * dt0 ^ (n - 2) = v2 ^ 2

/-- **Scale substitution.**  Replacing the time scale by `q·dt₀` multiplies the
relation by `q^(n−2)`, which is the framework's substitution rule
`vₙ·(q·dt₀)^(n−2) = v₂²·q^(n−2)`; the parameter `q = m_ℓ/m₀` is what links the
muonic and tauonic regimes. -/
theorem scale_substitution (n : ℕ) {vn dt0 q v2 : ℝ}
(h : vn * dt0 ^ (n - 2) = v2 ^ 2) :
vn * (q * dt0) ^ (n - 2) = v2 ^ 2 * q ^ (n - 2) := by
rw [mul_pow, ← h]
ring

/-! ## The velocity law -/

/-- Ejection velocity of particles produced in lepton–lepton collisions in the
generalized framework: the non-negative root of `vf² = v₂²·r^(n−2)`, where `r`
is the mass ratio `m_ℓ/m₀`. -/
def generalisedVelocity (n : ℕ) (v₂ r : ℝ) : ℝ := Real.sqrt (v₂ ^ 2 * r ^ (n - 2))

/-- **The velocity law**, for arbitrary generalized dimension `n`:
`vf² = v₂² · (m_ℓ/m₀)^(n−2)`. -/
theorem velocity_law (n : ℕ) (v₂ r : ℝ) (hr : 0 ≤ r) :
generalisedVelocity n v₂ r ^ 2 = v₂ ^ 2 * r ^ (n - 2) := by
unfold generalisedVelocity
exact Real.sq_sqrt (mul_nonneg (sq_nonneg v₂) (pow_nonneg hr _))

/-- The paper's form of the law, `vf = v₂ · √((m_ℓ/m₀)^(n−2))`, agrees with the
definition for non-negative input velocity. -/
theorem generalisedVelocity_eq_mul (n : ℕ) {v₂ r : ℝ} (hv : 0 ≤ v₂) :
generalisedVelocity n v₂ r = v₂ * Real.sqrt (r ^ (n - 2)) := by
unfold generalisedVelocity
rw [Real.sqrt_mul (sq_nonneg v₂), Real.sqrt_sq hv]

/-- Ejection velocities are non-negative. -/
theorem generalisedVelocity_nonneg (n : ℕ) (v₂ r : ℝ) : 0 ≤ generalisedVelocity n v₂ r :=
Real.sqrt_nonneg _

/-- Two-sided bounds on an ejection velocity, from exact bounds on `vf²`.
This is the workhorse that turns a computed square into a velocity interval
without any rounding step. -/
theorem generalisedVelocity_bounds {n : ℕ} {v₂ r a b : ℝ} (hb : 0 ≤ b)
(h₁ : a ^ 2 ≤ v₂ ^ 2 * r ^ (n - 2)) (h₂ : v₂ ^ 2 * r ^ (n - 2) ≤ b ^ 2) :
a ≤ generalisedVelocity n v₂ r ∧ generalisedVelocity n v₂ r ≤ b := by
unfold generalisedVelocity
exact ⟨Real.le_sqrt_of_sq_le h₁, (Real.sqrt_le_left hb).2 h₂⟩

/-- The velocity law at the framework's generalized dimension:
`vf² = v₂² · (m_ℓ/m₀)³`. -/
theorem velocity_law_five (v₂ r : ℝ) (hr : 0 ≤ r) :
generalisedVelocity generalizedDimension v₂ r ^ 2 = v₂ ^ 2 * r ^ 3 := by
rw [velocity_law generalizedDimension v₂ r hr, generalizedDimension_eq_five]

/-- The law's explicit linear-in-`v₂` form at `n = 5`: `vf = v₂·√(r³)`. -/
theorem generalisedVelocity_five_eq_mul {v₂ r : ℝ} (hv : 0 ≤ v₂) :
generalisedVelocity generalizedDimension v₂ r = v₂ * Real.sqrt (r ^ 3) := by
rw [generalisedVelocity_eq_mul generalizedDimension hv, generalizedDimension_eq_five]

/-- The ejection velocity grows with the lepton mass ratio: heavier collision
partners eject faster. -/
theorem generalisedVelocity_mono {n : ℕ} {v₂ r₁ r₂ : ℝ} (hr₁ : 0 ≤ r₁) (h : r₁ ≤ r₂) :
generalisedVelocity n v₂ r₁ ≤ generalisedVelocity n v₂ r₂ := by
unfold generalisedVelocity
apply Real.sqrt_le_sqrt
exact mul_le_mul_of_nonneg_left (pow_le_pow_left₀ hr₁ h _) (sq_nonneg v₂)

/-- The ejection velocity grows with the collision velocity. -/
theorem generalisedVelocity_mono_v₂ {n : ℕ} {v₂ v₂' r : ℝ} (hv : 0 ≤ v₂) (h : v₂ ≤ v₂')
(hr : 0 ≤ r) :
generalisedVelocity n v₂ r ≤ generalisedVelocity n v₂' r := by
unfold generalisedVelocity
apply Real.sqrt_le_sqrt
exact mul_le_mul_of_nonneg_right (pow_le_pow_left₀ hv h 2) (pow_nonneg hr _)

/-- The ejection velocity vanishes with the collision velocity: no collisions,
no propulsion. -/
theorem generalisedVelocity_zero (n : ℕ) (r : ℝ) : generalisedVelocity n 0 r = 0 := by
unfold generalisedVelocity
simp

end

end TheoreticalFramework
```

TheoreticalFramework/Velocity.lean

```
/-
# TheoreticalFramework.Velocity

Velocity scaling law, amplification factors and ejection velocities of the
muonic and tauonic regimes, with every physical constant bounded rigorously.

Units.  The framework states collision velocities in km/h and ejection
velocities in m/s and km/s.  All conversions are proved; no numeric value is
asserted without a surrounding interval, and each interval is closed by an exact
computation of `vf²` together with `generalisedVelocity_bounds`.

The two amplification factors of the framework are
`(m_μ/m_e)^(n−2) = 207³ = 8 869 743 ≈ 8.9 × 10⁶` and
`(m_τ/m_e)^(n−2) = 3477³ = 42 035 292 333 ≈ 4.2 × 10¹⁰`.

The muonic ceiling reproduces the source's `≈ 20 700 km/s`.  The tauonic ceiling
does *not* reproduce the source's `1.34 × 10¹² m/s`: the exact law gives
`4.2403 × 10¹²`, and the discrepancy is localised in
`source_tauon_max_velocity_underestimates` — the source evaluates
`√(1.80 × 10²⁵)` as `1.34 × 10¹²` where it is `4.24 × 10¹²`, a `√10` slip in the
decimal exponent.  The published figure is kept as a separate definition so that
the two can be compared without either being hidden.
-/
import TheoreticalFramework.Dimensions

namespace TheoreticalFramework

noncomputable section

/-! ## Units and framework constants -/

/-- Kilometres per hour to metres per second, `1 km/h = 5/18 m/s`. -/
def kmhToMs (v : ℝ) : ℝ := v * 5 / 18

/-- The conversion agrees with the definitional `1000/3600`. -/
theorem kmhToMs_eq (v : ℝ) : kmhToMs v = v * 1000 / 3600 := by
unfold kmhToMs
ring

/-- Standard gravity, `g₀ = 9.80665 m/s²`, used to express specific impulse. -/
def standardGravity : ℝ := 9.80665

theorem standardGravity_pos : 0 < standardGravity := by
norm_num [standardGravity]

/-- Speed of light in vacuum, m/s. -/
def speedOfLight : ℝ := 299792458

/-- Specific impulse, `Isp = v_e / g₀`, in seconds. -/
def specificImpulse (ve : ℝ) : ℝ := ve / standardGravity

/-- Vacuum specific impulse of a Raptor engine, seconds: the chemical benchmark
of the source text. -/
def raptorSpecificImpulse : ℝ := 363

/-! ## Amplification factors -/

/-- Upper end of the muonic input window, `v₂ = 25 000 km/h`. -/
def collisionVelocityMuonMax : ℝ := kmhToMs 25000

/-- The muonic input ceiling in m/s. -/
theorem collisionVelocityMuonMax_eq : collisionVelocityMuonMax = 62500 / 9 := by
unfold collisionVelocityMuonMax kmhToMs
norm_num

/-- `25 000 km/h ≈ 6 944 m/s`. -/
theorem collisionVelocityMuonMax_bounds :
6944 ≤ collisionVelocityMuonMax ∧ collisionVelocityMuonMax ≤ 6945 := by
rw [collisionVelocityMuonMax_eq]
constructor <;> norm_num

/-- Muonic amplification factor, `(m_μ/m_e)^(n−2)`. -/
def muonAmplification : ℝ := massRatioMuon ^ velocityExponent

/-- Tauonic amplification factor, `(m_τ/m_e)^(n−2)`. -/
def tauonAmplification : ℝ := massRatioTauon ^ velocityExponent

/-- `207³ = 8 869 743`. -/
theorem muonAmplification_eq : muonAmplification = 8869743 := by
unfold muonAmplification velocityExponent generalizedDimension casimirMassExponent
norm_num [massRatioMuon]

/-- `3477³ = 42 035 292 333`. -/
theorem tauonAmplification_eq : tauonAmplification = 42035292333 := by
unfold tauonAmplification velocityExponent generalizedDimension casimirMassExponent
norm_num [massRatioTauon]

/-- The muonic amplification is `(207)³ ≈ 8.9 × 10⁶`. -/
theorem muonAmplification_order :
(8.8 : ℝ) * 10 ^ 6 ≤ muonAmplification ∧ muonAmplification ≤ (8.9 : ℝ) * 10 ^ 6 := by
```

```
rw [muonAmplification_eq]
constructor <;> norm_num

/-- The tauonic amplification is `(3477)³ ≈ 4.2 × 10¹⁰`. -/
theorem tauonAmplification_order :
(4.2 : ℝ) * 10 ^ 10 ≤ tauonAmplification ∧ tauonAmplification ≤ (4.21 : ℝ) * 10 ^ 10 := by
rw [tauonAmplification_eq]
constructor <;> norm_num

/-- The tauonic amplification exceeds the muonic one by `(3477/207)³ ≈ 4.74 × 10³`. -/
theorem tauon_over_muon_amplification :
(4.73 : ℝ) * 10 ^ 3 ≤ tauonAmplification / muonAmplification ∧
tauonAmplification / muonAmplification ≤ (4.74 : ℝ) * 10 ^ 3 := by
rw [tauonAmplification_eq, muonAmplification_eq]
constructor
· rw [le_div_iff₀ (by norm_num)]
norm_num
· rw [div_le_iff₀ (by norm_num)]
norm_num

/-! ## Muonic regime: ejection velocity -/

/-- Maximum ejection velocity of the muonic regime, at `v₂ = 25 000 km/h`. -/
def muonMaxEjection : ℝ :=
generalisedVelocity generalizedDimension collisionVelocityMuonMax massRatioMuon

/-- The ejection velocity obeys the law: `vf² = v₂² · 207³`. -/
theorem muonMaxEjection_sq :
muonMaxEjection ^ 2 = collisionVelocityMuonMax ^ 2 * massRatioMuon ^ 3 := by
unfold muonMaxEjection
exact velocity_law_five _ _ massRatioMuon_pos.le

/-- `vf ≈ 2.06820 × 10⁷ m/s` for the muonic ceiling. -/
theorem muonMaxEjection_bounds :
(2.06820 : ℝ) * 10 ^ 7 ≤ muonMaxEjection ∧
muonMaxEjection ≤ (2.06821 : ℝ) * 10 ^ 7 := by
have h₁ : ((2.06820 : ℝ) * 10 ^ 7) ^ 2 ≤
collisionVelocityMuonMax ^ 2 * massRatioMuon ^ 3 := by
rw [collisionVelocityMuonMax_eq]
norm_num [massRatioMuon]
have h₂ : collisionVelocityMuonMax ^ 2 * massRatioMuon ^ 3 ≤
((2.06821 : ℝ) * 10 ^ 7) ^ 2 := by
rw [collisionVelocityMuonMax_eq]
norm_num [massRatioMuon]
have := generalisedVelocity_bounds (n := generalizedDimension)
(v₂ := collisionVelocityMuonMax) (r := massRatioMuon)
(a := (2.06820 : ℝ) * 10 ^ 7) (b := (2.06821 : ℝ) * 10 ^ 7)
(show (0 : ℝ) ≤ (2.06821 : ℝ) * 10 ^ 7 by norm_num) h₁ h₂
simpa [muonMaxEjection] using this

/-- The source's `≈ 20 700 km/s`, proved as an interval in km/s. -/
theorem muonMaxEjection_km_per_s :
(2.068 : ℝ) * 10 ^ 4 ≤ muonMaxEjection / 1000 ∧
muonMaxEjection / 1000 ≤ (2.07 : ℝ) * 10 ^ 4 := by
obtain ⟨h₁, h₂⟩ := muonMaxEjection_bounds
constructor
· rw [le_div_iff₀ (by norm_num)]
nlinarith
· rw [div_le_iff₀ (by norm_num)]
nlinarith

/-- The muonic ejection velocity is subluminal: the muonic regime is compatible
with ordinary special relativity. -/
theorem muonMaxEjection_subluminal : muonMaxEjection < speedOfLight := by
have := muonMaxEjection_bounds.2
unfold speedOfLight
linarith

/-- The muonic input window is subluminal as well. -/
theorem collisionVelocityMuonMax_subluminal : collisionVelocityMuonMax < speedOfLight := by
have := collisionVelocityMuonMax_bounds.2
unfold speedOfLight
linarith

/-- Specific impulse of the muonic ceiling, `Isp = v_e/g₀ ≈ 2.109 × 10⁶ s`. -/
theorem muonSpecificImpulse_bounds :
(2.108 : ℝ) * 10 ^ 6 ≤ specificImpulse muonMaxEjection ∧
specificImpulse muonMaxEjection ≤ (2.109 : ℝ) * 10 ^ 6 := by
obtain ⟨h₁, h₂⟩ := muonMaxEjection_bounds
have hg := standardGravity_pos
constructor
· rw [specificImpulse, le_div_iff₀ hg]
unfold standardGravity at hg ⊢
nlinarith
· rw [specificImpulse, div_le_iff₀ hg]
unfold standardGravity at hg ⊢
nlinarith

/-- The muonic impulse exceeds the Raptor benchmark by a factor of about
`5.8 × 10³`, i.e. "roughly 6 000 times" as the source states. -/
theorem muonImpulse_over_raptor :
(5.80 : ℝ) * 10 ^ 3 ≤ specificImpulse muonMaxEjection / raptorSpecificImpulse ∧
specificImpulse muonMaxEjection / raptorSpecificImpulse ≤ (5.82 : ℝ) * 10 ^ 3 := by
obtain ⟨h₁, h₂⟩ := muonSpecificImpulse_bounds
```

```
unfold raptorSpecificImpulse
constructor
· rw [le_div_iff₀ (by norm_num)]
nlinarith
· rw [div_le_iff₀ (by norm_num)]
nlinarith

/-! ## Tauonic regime: ejection velocity -/

/-- Input ceiling of the tauonic regime: the muonic regime's maximum ejection
velocity, `≈ 20 682 km/s`.  Taking the unrounded value is what makes the two
regimes exactly continuous. -/
def tauonicInputMax : ℝ := muonMaxEjection

/-- Maximum ejection velocity of the tauonic regime, at `v₂ = tauonicInputMax`. -/
def tauonMaxEjection : ℝ :=
generalisedVelocity generalizedDimension tauonicInputMax massRatioTauon

/-- The tauonic ejection velocity obeys the law: `vf² = v₂² · 3477³`. -/
theorem tauonMaxEjection_sq :
tauonMaxEjection ^ 2 = tauonicInputMax ^ 2 * massRatioTauon ^ 3 := by
unfold tauonMaxEjection
exact velocity_law_five _ _ massRatioTauon_pos.le

/-- `vf ≈ 4.2403 × 10¹²` m/s` for the tauonic ceiling. -/
theorem tauonMaxEjection_bounds :
(4.2403 : ℝ) * 10 ^ 12 ≤ tauonMaxEjection ∧
tauonMaxEjection ≤ (4.2404 : ℝ) * 10 ^ 12 := by
have hb := muonMaxEjection_bounds
have hsq₁ : ((2.06820 : ℝ) * 10 ^ 7) ^ 2 ≤ tauonicInputMax ^ 2 := by
rw [tauonicInputMax]
exact pow_le_pow_left₀ (show (0 : ℝ) ≤ (2.06820 : ℝ) * 10 ^ 7 by norm_num) hb.1 2
have hsq₂ : tauonicInputMax ^ 2 ≤ ((2.06821 : ℝ) * 10 ^ 7) ^ 2 := by
rw [tauonicInputMax]
exact pow_le_pow_left₀ (by linarith [hb.1] : (0 : ℝ) ≤ muonMaxEjection) hb.2 2
have hp₁ := mul_le_mul_of_nonneg_right hsq₁
(show (0 : ℝ) ≤ massRatioTauon ^ 3 by norm_num [massRatioTauon])
have hp₂ := mul_le_mul_of_nonneg_right hsq₂
(show (0 : ℝ) ≤ massRatioTauon ^ 3 by norm_num [massRatioTauon])
have h₁ : ((4.2403 : ℝ) * 10 ^ 12) ^ 2 ≤ tauonicInputMax ^ 2 * massRatioTauon ^ 3 := by
refine le_trans ?_ hp₁
norm_num [massRatioTauon]
have h₂ : tauonicInputMax ^ 2 * massRatioTauon ^ 3 ≤ ((4.2404 : ℝ) * 10 ^ 12) ^ 2 := by
refine le_trans hp₂ ?_
norm_num [massRatioTauon]
have := generalisedVelocity_bounds (n := generalizedDimension)
(v₂ := tauonicInputMax) (r := massRatioTauon)
(a := (4.2403 : ℝ) * 10 ^ 12) (b := (4.2404 : ℝ) * 10 ^ 12)
(show (0 : ℝ) ≤ (4.2404 : ℝ) * 10 ^ 12 by norm_num) h₁ h₂
simpa [tauonMaxEjection] using this

/-- The tauonic ceiling is superluminal: the framework's rejection of the
Lorentz factor is not optional decoration, it is what its own numbers require. -/
theorem tauonMaxEjection_superluminal : speedOfLight < tauonMaxEjection := by
have := tauonMaxEjection_bounds.1
unfold speedOfLight
linarith

/-- The tauonic input window is subluminal; only the output exceeds `c`. -/
theorem tauonicInputMax_subluminal : tauonicInputMax < speedOfLight := by
rw [tauonicInputMax]
exact muonMaxEjection_subluminal

/-! ## The tauonic figures of the source text

The source states `vf ≈ 1.34 × 10¹²` m/s` at the tauonic ceiling, and
`vf ≈ 9.0 × 10¹⁰` m/s` at a median input of `500 000 km/h`.  The exact law gives
`4.2403 × 10¹²` and `2.8477 × 10¹⁰`.  In both cases the printed value is `√10`
too small: the source's own intermediate `vf² = 1.80 × 10²⁵ m²/s²` is correct,
but its square root is taken as `1.34 × 10¹²` instead of `4.24 × 10¹²`.  The
discrepancy is proved here rather than asserted, and both values are kept so
that a reader can see exactly where the two diverge.
-/

/-- The tauonic ceiling ejection velocity as printed in the source text. -/
def sourceTauonMaxVelocity : ℝ := 1.34 * 10 ^ 12

/-- The tauonic median ejection velocity as printed in the source text. -/
def sourceTauonMedianVelocity : ℝ := 9.0 * 10 ^ 10

/-- The source figure, squared, falls short of the law by a factor of `10.01`:
the printed value is `√10` too small. -/
theorem source_tauon_max_velocity_underestimates :
(10.01 : ℝ) ≤ tauonMaxEjection ^ 2 / sourceTauonMaxVelocity ^ 2 ∧
tauonMaxEjection ^ 2 / sourceTauonMaxVelocity ^ 2 ≤ (10.02 : ℝ) := by
obtain ⟨h₁, h₂⟩ := tauonMaxEjection_bounds
have hsq₁ : ((4.2403 : ℝ) * 10 ^ 12) ^ 2 ≤ tauonMaxEjection ^ 2 :=
pow_le_pow_left₀ (show (0 : ℝ) ≤ (4.2403 : ℝ) * 10 ^ 12 by norm_num) h₁ 2
have hsq₂ : tauonMaxEjection ^ 2 ≤ ((4.2404 : ℝ) * 10 ^ 12) ^ 2 :=
pow_le_pow_left₀ (by linarith : (0 : ℝ) ≤ tauonMaxEjection) h₂ 2
unfold sourceTauonMaxVelocity
constructor
· rw [le_div_iff₀ (by norm_num)]
```

```
nlinarith
· rw [div_le_iff₀ (by norm_num)]
nlinarith

/-- Consequently the source figure underestimates the ejection velocity itself by
the factor `√10 ≈ 3.164`. -/
theorem source_tauon_max_velocity_factor :
(3.164 : ℝ) ≤ tauonMaxEjection / sourceTauonMaxVelocity ∧
tauonMaxEjection / sourceTauonMaxVelocity ≤ (3.165 : ℝ) := by
obtain ⟨h₁, h₂⟩ := tauonMaxEjection_bounds
unfold sourceTauonMaxVelocity
constructor
· rw [le_div_iff₀ (by norm_num)]
nlinarith
· rw [div_le_iff₀ (by norm_num)]
nlinarith

/-- Tauonic ejection velocity at the entry velocity of the tauonic window. -/
def tauonEntryEjection : ℝ :=
generalisedVelocity generalizedDimension collisionVelocityMuonMax massRatioTauon

/-- `vf ≈ 1.4238 × 10⁹ m/s` at the tauonic entry velocity: the source's
`1.42 × 10⁹` is correct here, because this `vf²` happens to carry an even
decimal exponent. -/
theorem tauonEntryEjection_bounds :
(1.4237 : ℝ) * 10 ^ 9 ≤ tauonEntryEjection ∧
tauonEntryEjection ≤ (1.4239 : ℝ) * 10 ^ 9 := by
have h₁ : ((1.4237 : ℝ) * 10 ^ 9) ^ 2 ≤
collisionVelocityMuonMax ^ 2 * massRatioTauon ^ 3 := by
rw [collisionVelocityMuonMax_eq]
norm_num [massRatioTauon]
have h₂ : collisionVelocityMuonMax ^ 2 * massRatioTauon ^ 3 ≤
((1.4239 : ℝ) * 10 ^ 9) ^ 2 := by
rw [collisionVelocityMuonMax_eq]
norm_num [massRatioTauon]
have := generalisedVelocity_bounds (n := generalizedDimension)
(v₂ := collisionVelocityMuonMax) (r := massRatioTauon)
(a := (1.4237 : ℝ) * 10 ^ 9) (b := (1.4239 : ℝ) * 10 ^ 9)
(show (0 : ℝ) ≤ (1.4239 : ℝ) * 10 ^ 9 by norm_num) h₁ h₂
simpa [tauonEntryEjection] using this

/-- Tauonic ejection velocity at a median input of `500 000 km/h`. -/
def tauonMedianEjection : ℝ :=
generalisedVelocity generalizedDimension (kmhToMs 500000) massRatioTauon

/-- `vf ≈ 2.8477 × 10¹⁰ m/s` at the median input — not the `9.0 × 10¹⁰` printed
in the source, which carries the same `√10` slip. -/
theorem tauonMedianEjection_bounds :
(2.8475 : ℝ) * 10 ^ 10 ≤ tauonMedianEjection ∧
tauonMedianEjection ≤ (2.8476 : ℝ) * 10 ^ 10 := by
have hvm : kmhToMs 500000 = 1250000 / 9 := by
unfold kmhToMs
norm_num
have h₁ : ((2.8475 : ℝ) * 10 ^ 10) ^ 2 ≤
kmhToMs 500000 ^ 2 * massRatioTauon ^ 3 := by
rw [hvm]
norm_num [massRatioTauon]
have h₂ : kmhToMs 500000 ^ 2 * massRatioTauon ^ 3 ≤
((2.8476 : ℝ) * 10 ^ 10) ^ 2 := by
rw [hvm]
norm_num [massRatioTauon]
have := generalisedVelocity_bounds (n := generalizedDimension)
(v₂ := kmhToMs 500000) (r := massRatioTauon)
(a := (2.8475 : ℝ) * 10 ^ 10) (b := (2.8476 : ℝ) * 10 ^ 10)
(show (0 : ℝ) ≤ (2.8476 : ℝ) * 10 ^ 10 by norm_num) h₁ h₂
simpa [tauonMedianEjection] using this

end

end TheoreticalFramework
```

TheoreticalFramework/Regimes.lean

```
/-
# TheoreticalFramework.Regimes

The two nested propulsive regimes, their continuity at the domain boundary, the
structural algebra of the amplification factor, and an exact treatment of what
the framework does *not* fix: thrust.

Structure of the argument.

* A `PropulsiveRegime` is a mass ratio together with the window of admissible
collision velocities `v₂` and the law-induced output.  Both regimes are
instances, so everything proved about `PropulsiveRegime` applies to both.
* The muonic window is `]0, 25 000 km/h]`; the tauonic window runs from
`25 000 km/h` to the muonic regime's *output* ceiling, `≈ 20 682 km/s`.
Hence the windows tile one interval (`regimes_tile`), the tauonic window is
nested inside the muonic output range (`tauonic_window_nested_in_output`), and
the output ceiling of the first regime is the input ceiling of the second
(`chain_closes`).  That is the exact content of the continuity claim.
* The amplification `r ↦ r^(n−2)` is a multiplicative character
(`amplification_mul`), so the tauonic factor factors through the muonic one.
* Thrust is *not* determined by the geometry.  The kinematics fix the exhaust
velocity and the thrust-to-power ratio `T/P = 2/v_e` exactly; the absolute
thrust is free (`thrust_not_determined`).  What remains is an engineering
question, recorded as `ThrustOptimizationConjecture` and left unproven, which
is what the framework claims and no more.
-/
import TheoreticalFramework.Velocity

set_option linter.unusedVariables false

namespace TheoreticalFramework

noncomputable section

/-! ## Amplification as a multiplicative character -/

/-- The amplification factor is multiplicative in the mass ratio: chaining two
mass scales multiplies the amplification.  This is what makes the regimes
composable rather than merely analogous. -/
theorem amplification_mul (r₁ r₂ : ℝ) :
(r₁ * r₂) ^ velocityExponent = r₁ ^ velocityExponent * r₂ ^ velocityExponent :=
mul_pow r₁ r₂ _

/-- The amplification is strictly increasing in the mass ratio on the positive
axis: heavier collision partners amplify more. -/
theorem amplification_strictMono :
StrictMonoOn (fun r : ℝ => r ^ velocityExponent) (Set.Ioi 0) := by
intro r₁ hr₁ r₂ hr₂ hlt
exact pow_lt_pow_left₀ hlt (le_of_lt hr₁) (by decide)

/-- The tauonic amplification is the muonic one times the amplification of the
residual ratio `m_τ/m_μ`: `3477³ = 207³ · (3477/207)³`. -/
theorem tauon_amplification_factors :
tauonAmplification = muonAmplification * (massRatioTauon / massRatioMuon) ^ velocityExponent := by
unfold tauonAmplification muonAmplification
rw [div_pow, mul_div_cancel₀ _ (pow_ne_zero _ (ne_of_gt massRatioMuon_pos))]

/-! ## The ejection velocity never falls below the collision velocity -/

/-- For a mass ratio at least one, the ejection velocity is at least the
collision velocity.  This is what lets the tauonic window start where the muonic
window ends. -/
theorem generalisedVelocity_ge_input {n : ℕ} {v₂ r : ℝ} (hr : 1 ≤ r) :
v₂ ≤ generalisedVelocity n v₂ r := by
have hone : (1 : ℝ) ≤ r ^ (n - 2) := one_le_pow₀ hr
have hsq : v₂ ^ 2 ≤ v₂ ^ 2 * r ^ (n - 2) := le_mul_of_one_le_right (sq_nonneg v₂) hone
exact Real.le_sqrt_of_sq_le hsq

theorem one_le_massRatioMuon : (1 : ℝ) ≤ massRatioMuon := by
norm_num [massRatioMuon]

theorem one_le_massRatioTauon : (1 : ℝ) ≤ massRatioTauon := by
norm_num [massRatioTauon]

/-! ## Regimes -/

/-- A propulsive regime of the framework: a charged-lepton mass ratio, the
window of admissible collision velocities, and the law-induced quantities
derived from them. -/
structure PropulsiveRegime where
/-- Display name of the regime. -/
name : String
/-- Colliding particle, for the record. -/
particle : String
/-- Mass ratio `m_ℓ/m_e` of the colliding leptons. -/
massRatio : ℝ
massRatio_pos : 0 < massRatio
/-- Lower end of the admissible input window, in m/s. -/
inputLower : ℝ
```

```
/-- Upper end of the admissible input window, in m/s. -/
inputUpper : ℝ
input_window : inputLower ≤ inputUpper

namespace PropulsiveRegime

/-- Amplification factor of the regime, `(m_ℓ/m_e)^(n−2)`. -/
def amplification (R : PropulsiveRegime) : ℝ := R.massRatio ^ velocityExponent

/-- Maximum ejection velocity of the regime, at the top of its input window. -/
def maxOutput (R : PropulsiveRegime) : ℝ :=
generalisedVelocity generalizedDimension R.inputUpper R.massRatio

/-- The input window as a set. -/
def window (R : PropulsiveRegime) : Set ℝ := Set.Icc R.inputLower R.inputUpper

/-- The regime's maximum output obeys the velocity law. -/
theorem maxOutput_sq (R : PropulsiveRegime) :
R.maxOutput ^ 2 = R.inputUpper ^ 2 * R.massRatio ^ 3 := by
unfold maxOutput
exact velocity_law_five _ _ (le_of_lt R.massRatio_pos)

/-- The maximum output is at least the input ceiling. -/
theorem inputUpper_le_maxOutput (R : PropulsiveRegime) (h : 1 ≤ R.massRatio) :
R.inputUpper ≤ R.maxOutput :=
generalisedVelocity_ge_input h

/-- Within a regime the velocity law is continuous in the collision velocity. -/
theorem continuous_maxOutput (R : PropulsiveRegime) :
Continuous fun v₂ : ℝ => generalisedVelocity generalizedDimension v₂ R.massRatio := by
unfold generalisedVelocity
exact Real.continuous_sqrt.comp (by fun_prop)

/-- Within a regime the velocity law is continuous in the mass ratio. -/
theorem continuous_in_massRatio (R : PropulsiveRegime) (v₂ : ℝ) :
Continuous fun r : ℝ => generalisedVelocity generalizedDimension v₂ r := by
unfold generalisedVelocity
refine Real.continuous_sqrt.comp ?_
exact continuous_const.mul (continuous_id.pow _)

end PropulsiveRegime

/-- **The muonic regime.**  Muon–muon collisions, mass ratio `207`, collision
velocities up to `25 000 km/h`; replaces chemical propellants. -/
def muonicRegime : PropulsiveRegime where
name := "muonic"
particle := "μ⁻/μ⁺"
massRatio := massRatioMuon
massRatio_pos := massRatioMuon_pos
inputLower := 0
inputUpper := collisionVelocityMuonMax
input_window := by
have h := collisionVelocityMuonMax_bounds.1
linarith

/-- **The tauonic regime.**  Tauon–tauon collisions, mass ratio `3477`, input
window from the muonic ceiling up to the muonic regime's output ceiling. -/
def tauonicRegime : PropulsiveRegime where
name := "tauonic"
particle := "τ⁻/τ⁺"
massRatio := massRatioTauon
massRatio_pos := massRatioTauon_pos
inputLower := collisionVelocityMuonMax
inputUpper := tauonicInputMax
input_window := by
have h := generalisedVelocity_ge_input
(n := generalizedDimension) (v₂ := collisionVelocityMuonMax) (r := massRatioMuon)
one_le_massRatioMuon
simpa [tauonicInputMax, muonMaxEjection] using h

/-- The muonic amplification, through the structure. -/
theorem muonicRegime_amplification :
muonicRegime.amplification = muonAmplification := by
unfold PropulsiveRegime.amplification muonicRegime muonAmplification
rfl

/-- The tauonic amplification, through the structure. -/
theorem tauonicRegime_amplification :
tauonicRegime.amplification = tauonAmplification := by
unfold PropulsiveRegime.amplification tauonicRegime tauonAmplification
rfl

/-- The regime hierarchy is strict: the tauonic amplification exceeds the
muonic one by `≈ 4.74 × 10³`. -/
theorem regimes_strictly_ordered :
muonicRegime.amplification < tauonicRegime.amplification := by
rw [muonicRegime_amplification, tauonicRegime_amplification,
muonAmplification_eq, tauonAmplification_eq]
norm_num

/-- The muonic regime's output ceiling, through the structure. -/
theorem muonicRegime_maxOutput : muonicRegime.maxOutput = muonMaxEjection := by
unfold PropulsiveRegime.maxOutput muonicRegime muonMaxEjection
```

```
rfl

/-- The tauonic regime's output ceiling, through the structure. -/
theorem tauonicRegime_maxOutput : tauonicRegime.maxOutput = tauonMaxEjection := by
unfold PropulsiveRegime.maxOutput tauonicRegime tauonMaxEjection
rfl

/-! ## Continuity at the domain boundary -/

/-- **Adjacency.**  The tauonic window begins exactly where the muonic window
ends: `25 000 km/h`, the muonic collision ceiling. -/
theorem windows_adjacent : muonicRegime.inputUpper = tauonicRegime.inputLower := rfl

/-- **The chain closes.**  The upper end of the tauonic *input* window is the
muonic regime's maximum *output*.  This is the boundary the framework calls
`v₂ = 20 700 km/s`, and taking the unrounded value is what makes the match
exact rather than approximate. -/
theorem chain_closes : tauonicRegime.inputUpper = muonicRegime.maxOutput := rfl

/-- **Tiling.**  The two input windows cover one interval with no gap and no
overlap. -/
theorem regimes_tile :
muonicRegime.window ∪ tauonicRegime.window
= Set.Icc (0 : ℝ) tauonicRegime.inputUpper := by
ext v
simp only [PropulsiveRegime.window, muonicRegime, tauonicRegime, Set.mem_union,
Set.mem_Icc]
constructor
· rintro (⟨h₀, h₁⟩ | ⟨h₁, h₂⟩)
· exact ⟨h₀, le_trans h₁ (by
have := generalisedVelocity_ge_input (n := generalizedDimension)
(v₂ := collisionVelocityMuonMax) (r := massRatioMuon) one_le_massRatioMuon
simpa [tauonicInputMax, muonMaxEjection] using this)⟩
· exact ⟨le_trans (by
have := collisionVelocityMuonMax_bounds
linarith [this.1]) h₁, h₂⟩
· rintro ⟨h₀, h₂⟩
rcases le_total v collisionVelocityMuonMax with h | h
· exact Or.inl ⟨h₀, h⟩
· exact Or.inr ⟨h, h₂⟩

/-- **Nesting.**  The tauonic window lies inside the muonic regime's output
range: the second regime's inputs are the first regime's reachable outputs. -/
theorem tauonic_window_nested_in_output :
tauonicRegime.window ⊆ Set.Ioc (0 : ℝ) muonicRegime.maxOutput := by
intro v hv
have h := regimes_tile
simp only [PropulsiveRegime.window, tauonicRegime, Set.mem_Icc] at hv
refine ⟨?_, ?_⟩
· have := collisionVelocityMuonMax_bounds
linarith [hv.1]
· rw [tauonicInputMax] at hv
exact hv.2

/-- **The domain boundary.**  The boundary of the source's §8.2, `v₂ = 20 700 km/s`,
is the tauonic input ceiling, and it is the muonic output ceiling unrounded. -/
def domainBoundary : ℝ := tauonicRegime.inputUpper

/-- The boundary is the muonic regime's maximum output. -/
theorem domainBoundary_eq_muonic_output : domainBoundary = muonicRegime.maxOutput := rfl

/-- The boundary is the muonic ejection ceiling, hence `≈ 20 700 km/s`. -/
theorem domainBoundary_km_per_s :
(2.068 : ℝ) * 10 ^ 4 ≤ domainBoundary / 1000 ∧
domainBoundary / 1000 ≤ (2.07 : ℝ) * 10 ^ 4 := by
simpa [domainBoundary, tauonicRegime, tauonicInputMax] using muonMaxEjection_km_per_s

/-- The boundary itself is subluminal; it is the tauonic *output* that exceeds
`c`, not the input ceiling that defines the hand-off. -/
theorem domainBoundary_subluminal : domainBoundary < speedOfLight := by
simpa [domainBoundary, tauonicRegime, tauonicInputMax] using muonMaxEjection_subluminal

/-- **Continuity of the two-regime structure**, collected in one statement: the
windows are adjacent, the chain closes on the muonic output ceiling, they tile
an interval, and the law is continuous in the collision velocity throughout. -/
theorem regime_continuity :
muonicRegime.inputUpper = tauonicRegime.inputLower ∧
tauonicRegime.inputUpper = muonicRegime.maxOutput ∧
muonicRegime.window ∪ tauonicRegime.window
= Set.Icc (0 : ℝ) tauonicRegime.inputUpper ∧
Continuous (fun v₂ : ℝ =>
generalisedVelocity generalizedDimension v₂ muonicRegime.massRatio) ∧
Continuous (fun v₂ : ℝ =>
generalisedVelocity generalizedDimension v₂ tauonicRegime.massRatio) :=
⟨windows_adjacent, chain_closes, regimes_tile,
muonicRegime.continuous_maxOutput, tauonicRegime.continuous_maxOutput⟩

/-! ## Thrust: what the framework fixes and what it does not

The kinematics fix the exhaust velocity.  Momentum and energy balance then fix
the thrust-to-power ratio exactly, `T/P = 2/v_e`, and nothing else: the absolute
thrust is set by mass flow and power, which are engineering parameters.  This is
the precise sense in which "empirical thrust optimisation" is outside the
```

```
framework, and it is proved rather than asserted.
-/

/-- A thruster design obeying the framework's momentum and energy balance. -/
structure ThrusterDesign where
/-- Exhaust velocity of the propulsive particles, m/s. -/
exhaustVelocity : ℝ
exhaustVelocity_pos : 0 < exhaustVelocity
/-- Propellant mass flow, kg/s. -/
massFlow : ℝ
massFlow_pos : 0 < massFlow
/-- Thrust, N: momentum flux `ṁ·v_e`. -/
thrust : ℝ
thrust_eq : thrust = massFlow * exhaustVelocity
/-- Jet power, W: kinetic energy flux `½·ṁ·v_e²`. -/
jetPower : ℝ
power_eq : jetPower = massFlow * exhaustVelocity ^ 2 / 2

/-- Vacuum exhaust velocity of the Raptor benchmark. -/
def raptorExhaustVelocity : ℝ := raptorSpecificImpulse * standardGravity

/-- **The thrust-power tradeoff.**  For any design, `T/P = 2/v_e` exactly.  High
specific impulse costs thrust per watt, without exception. -/
theorem thrust_to_power (d : ThrusterDesign) :
d.thrust / d.jetPower = 2 / d.exhaustVelocity := by
rw [d.thrust_eq, d.power_eq]
field_simp [ne_of_gt d.massFlow_pos, ne_of_gt d.exhaustVelocity_pos]

/-- Thrust in terms of jet power and exhaust velocity. -/
theorem thrust_eq_power (d : ThrusterDesign) :
d.thrust = 2 * d.jetPower / d.exhaustVelocity := by
rw [d.thrust_eq, d.power_eq]
field_simp [ne_of_gt d.exhaustVelocity_pos]

/-- At the muonic ceiling, thrust per watt is `≈ 9.67 × 10⁻⁸ N/W`. -/
theorem muonic_thrust_to_power_bounds (d : ThrusterDesign)
(h : d.exhaustVelocity = muonMaxEjection) :
(96 : ℝ) / 10 ^ 9 ≤ d.thrust / d.jetPower ∧
d.thrust / d.jetPower ≤ (97 : ℝ) / 10 ^ 9 := by
rw [thrust_to_power, h]
obtain ⟨h₁, h₂⟩ := muonMaxEjection_bounds
constructor
· rw [le_div_iff₀ (by linarith)]
nlinarith
· rw [div_le_iff₀ (by linarith)]
nlinarith

/-- The muonic exhaust velocity is `≈ 5.8 × 10³` times the Raptor exhaust
velocity, so its thrust per watt is that many times *worse*: the specific-impulse
gain and the thrust-per-watt loss are the same number. -/
theorem muonic_thrust_to_power_shortfall :
(5.80 : ℝ) * 10 ^ 3 ≤ muonMaxEjection / raptorExhaustVelocity ∧
muonMaxEjection / raptorExhaustVelocity ≤ (5.82 : ℝ) * 10 ^ 3 := by
have hrv : raptorExhaustVelocity = 363 * 9.80665 := by
unfold raptorExhaustVelocity raptorSpecificImpulse standardGravity
norm_num
obtain ⟨h₁, h₂⟩ := muonMaxEjection_bounds
constructor
· rw [hrv, le_div_iff₀ (by norm_num)]
nlinarith
· rw [hrv, div_le_iff₀ (by norm_num)]
nlinarith

/-- **Thrust is not determined by the geometry.**  At the exhaust velocity the
leptonic kinematics fix, any positive thrust is attainable by a suitable design:
thrust is bought with mass flow and power, not with the velocity law. -/
theorem thrust_not_determined (T : ℝ) (hT : 0 < T) :
∃ d : ThrusterDesign, d.exhaustVelocity = muonMaxEjection ∧ d.thrust = T := by
have hv : 0 < muonMaxEjection := by
have := muonMaxEjection_bounds.1
linarith
refine ⟨{ exhaustVelocity := muonMaxEjection, exhaustVelocity_pos := hv,
massFlow := T / muonMaxEjection, massFlow_pos := div_pos hT hv,
thrust := T, thrust_eq := ?_, jetPower := T * muonMaxEjection / 2,
power_eq := ?_ }, rfl, rfl⟩
· field_simp
· field_simp

/-- The engineering question the framework leaves open: at the exhaust velocity
the geometry fixes, is there a design reaching a thrust target `T` within a
power budget `P`?  Nothing in the geometric development decides it — see
`thrust_not_determined` — so it is recorded, not proved. -/
def ThrustOptimizationConjecture (T P : ℝ) : Prop :=
∃ d : ThrusterDesign, d.exhaustVelocity = muonMaxEjection ∧ d.thrust = T ∧ d.jetPower ≤ P

/-- The conjecture is not vacuous: a large enough power budget always suffices,
because the thrust-power relation is a hyperbola and nothing else constrains it. -/
theorem thrust_optimization_satisfiable (T : ℝ) (hT : 0 < T) :
ThrustOptimizationConjecture T (T * muonMaxEjection / 2 + 1) := by
obtain ⟨d, hv, hth⟩ := thrust_not_determined T hT
refine ⟨d, hv, hth, ?_⟩
have hvpos : 0 < muonMaxEjection := by
have := muonMaxEjection_bounds.1
```

```
linarith
have hmv : d.massFlow * d.exhaustVelocity = T := by
rw [← d.thrust_eq, hth]
rw [hv] at hmv
have hmf : d.massFlow = T / muonMaxEjection := by
rw [eq_div_iff (ne_of_gt hvpos)]
linarith
have hp := d.power_eq
rw [hv, hmf] at hp
rw [hp]
have hnorm : T / muonMaxEjection * muonMaxEjection ^ 2 / 2 = T * muonMaxEjection / 2 := by
field_simp
rw [hnorm]
linarith

/-- Consequently the thrust question is *empirical*, not geometric: the
framework fixes `v_e` and hence `T/P`, and leaves `(T, P)` free along that
hyperbola.  This is the formal statement of "empirical thrust optimisations
remain unproven". -/
theorem thrust_optimization_is_empirical (T P : ℝ) (hT : 0 < T) (hP : 0 < P)
(hpower : T ≤ 2 * P / muonMaxEjection) :
ThrustOptimizationConjecture T P := by
have hv : 0 < muonMaxEjection := by
have := muonMaxEjection_bounds.1
linarith
refine ⟨{ exhaustVelocity := muonMaxEjection, exhaustVelocity_pos := hv,
massFlow := T / muonMaxEjection, massFlow_pos := div_pos hT hv,
thrust := T, thrust_eq := ?_, jetPower := T * muonMaxEjection / 2,
power_eq := ?_ }, rfl, rfl, ?_⟩
· field_simp
· field_simp
· have hp : T * muonMaxEjection ≤ 2 * P := by
have h' := hpower
rw [le_div_iff₀ hv] at h'
linarith
show T * muonMaxEjection / 2 ≤ P
rw [div_le_iff₀ (by norm_num : (0 : ℝ) < 2)]
linarith

end

end TheoreticalFramework
```